

Spacing statistics for a point scatterer on the cubic three-torus

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Abstract

We study the new eigenvalues of a fixed point interaction on the cubic three-torus. Their mean-one consecutive spacings converge to a probability law independent of the interaction parameter. We characterize this law by the consecutive zeros of a random meromorphic function built from three-squares congruence densities. Its small-gap distribution has the form $s^5 \Psi(\log_2(s^{-2})) + o(s^5)$, where Ψ is continuous, positive, and one-periodic. The proof combines periodic mean-square approximation, a weighted Hilbert-transform estimate, and Fourier estimates that retain the interactions between primes.

1 Introduction

A central question in quantum chaos is: how does the spectral information of a quantum system reflect the geometric and dynamical properties of the corresponding classical system? For instance, the Berry–Tabor [BT] and Bohigas–Giannoni–Schmit [BGS] conjectures predict that generic integrable systems have Poisson local statistics of their quantum energy levels, whereas generic chaotic systems have random matrix statistics, with the ensemble determined by the symmetries. These conjectures motivated the study of explicit models where something can be said about the quantum energy levels. One notable example is the Šeba billiard [S], and its toral analogue, a point scatterer on a flat torus. The spectrum retains the Laplace *old eigenvalues*, with each multiplicity reduced by one, and contains a sequence of *new eigenvalues*. The local statistics of these new eigenvalues have been closely studied [RU, K]. In this article, we give a limiting gap distribution for the new spectrum and prove a sharp asymptotic for the frequency of small gaps.

Let $\mathbb{T}^3 := \mathbb{R}^3 / (2\pi\mathbb{Z})^3$ denote the three-torus. The positive Laplace eigenvalues are the integers $m > 0$ such that $r_3(m) > 0$, where

$$r_3(m) := \#\{v \in \mathbb{Z}^3 : |v|^2 = m\}.$$

The multiplicity at m is $r_3(m)$. Fix a point $o \in \mathbb{T}^3$. The Hamiltonian with a point scatterer is formally written as

$$H_{\alpha,o} := -\Delta + \alpha\delta_o.$$

It is defined rigorously through a nontrivial self-adjoint extension of the Laplacian restricted away from o . We use $c \in \mathbb{R}$ for the fixed extension parameter in the spectral equation (2.1). The formal coupling α will not be used below.

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Between every two consecutive distinct old eigenvalues there is exactly one new eigenvalue. Thus the new eigenvalues *interlace* the old ones. Write

$$\Lambda_1(c) < \Lambda_2(c) < \cdots$$

for the positive new eigenvalues in increasing order. The new spectrum has density $\rho := 5/6$, as recalled in Section 2.2. Its mean-one spacings and empirical spacing measure are

$$s_j(c) := \frac{5}{6}(\Lambda_{j+1}(c) - \Lambda_j(c)), \quad \mu_N^{(c)} := \frac{1}{N} \sum_{j=1}^N \delta_{s_j(c)}.$$

Theorem 1.1. *There is a probability measure μ on $[0, 10/3]$ such that, for every fixed $c \in \mathbb{R}$,*

$$\mu_N^{(c)} \implies \mu \quad (N \rightarrow \infty). \quad (1.1)$$

The limit is independent of c , has mean one, and satisfies $\mu(\{0\}) = 0$.

The convergence in (1.1) requires no averaging over the interaction parameter. In particular, at every continuity point s of $F(s) := \mu([0, s])$,

$$\frac{1}{N} \# \{j \leq N : s_j(c) \leq s\} \longrightarrow F(s).$$

An explicit description of μ is given in (6.1). The bounded support comes from interlacing.

Our second result concerns the mass that this law assigns to small gaps.

Theorem 1.2. *There is a continuous, strictly positive, one-periodic function Ψ such that*

$$F(s) = s^5 \Psi(\log_2(s^{-2})) + o(s^5) \quad (s \downarrow 0). \quad (1.2)$$

In particular, $F(s) \asymp s^5$.

This gives a strong form of level repulsion: the limiting law assigns mass of order s^5 to $[0, s]$. For comparison, the cumulative Poisson spacing law is of order s , and the GOE law is of order s^2 near zero. The periodic coefficient reflects the powers of four in the three-squares problem.

1.1 Relation to earlier work

The model originates with Šeba [S], who studied a rectangular billiard with a point interaction and observed spectral and eigenfunction behavior associated with quantum chaos, despite the essentially integrable classical motion. Shigehara [Sh] clarified the role of the coupling: in two dimensions, the regime displaying repulsion requires the extension parameter to vary with energy. A fixed extension has a different high-energy limit. The toral model replaces the billiard boundary by periodic boundary conditions.

Rudnick–Ueberschär [RU] made this distinction precise using a trace formula. In two dimensions and at fixed coupling, the new levels approach the distinct old levels on the mean-spacing scale. Consequently, their limiting spacing distributions agree whenever either exists. In three dimensions, they proved instead that the mean old-to-new displacement is asymptotic to half the mean old spacing. This provides information about the average displacement without determining the spacing distribution.

For the cubic three-torus, Kurlberg [K] proved that the upper asymptotic proportion of gaps at most s is $O_\varepsilon(s^{4-\varepsilon})$ for every $\varepsilon > 0$. This bound does not assume existence of a limiting spacing law. His argument relates small gaps to unusually small normalized multiplicities and identifies divisibility by large powers of four as the principal arithmetic mechanism. Kurlberg also discusses a sixth-order heuristic; the fifth-order law here comes from analyzing the regular Green value together with the central multiplicity and the slope.

Alongside this work on the distribution of the eigenvalues, the distribution and shape of the new eigenfunctions have also been studied. Kurlberg–Ueberschär proved quantum ergodicity on arithmetic tori and, in contrast, established superscars on Diophantine irrational tori [KU1, KU2].

Our proof uses comparison with an arithmetic probability model which is in the spirit of Kurlberg–Lester [KL], who proved Poisson angular spacings for lattice points on circles along a density-one sequence of admissible radii. Their observable and limiting law are different. In the present problem, the congruence model preserves the dependence between neighboring multiplicities.

1.2 Proof outline

There are two separate parts. For convergence, put

$$a(n) := \frac{r_3(n)}{2\pi\sqrt{n}}, \quad P_M(u) := M^{-2} \#\{x \bmod M : |x|^2 \equiv u \pmod{M}\}.$$

The second-moment formula of Choi–Kumchev–Osburn [CKO] gives, with $M_Q := \text{lcm}(1, \dots, Q)$,

$$\lim_{X \rightarrow \infty} \frac{1}{X} \sum_{n \leq X} |a(n) - P_{M_Q}(n)|^2 \ll Q^{-1}.$$

The new eigenvalues solve $G(\lambda) = c$, where

$$G(\lambda) := \sum_{n \geq 0} r_3(n) \left(\frac{1}{n - \lambda} - \frac{n}{n^2 + 1} \right)$$

is the regularized scalar Green function. A weighted discrete Hilbert transform transfers the approximation of a to this spectral equation:

$$\frac{G(m+z) - c}{2\pi\sqrt{m}} \approx -\frac{\pi}{M_Q} \sum_{b \bmod M_Q} P_{M_Q}(m+b) \cot \frac{\pi(z-b)}{M_Q}.$$

We take the limit on the profinite integers. Its surviving poles agree almost surely with the three-squares criterion; consequently its two zeros adjacent to the central pole determine the limiting spacing law.

We next explain the small-gap argument. Fix a realization H of the limiting Green function for which the origin is a pole. Let $-u < 0 < v$ be the two zeros adjacent to this pole, and put $d := u + v$. Thus d is the gap whose distribution we study. Near the origin we write

$$H(z) = -\frac{R}{z} + h + Dz + O(Dz^2),$$

where $R > 0$ is the magnitude of the residue, h is the value at the origin of the regular part of H , and $D > 0$ is the derivative of the regular part. Solving the corresponding quadratic equation gives

$$d = q + O(q^2), \quad q := \frac{\sqrt{h^2 + 4DR}}{D}.$$

Consequently, a gap of size at most s requires both

$$R \ll Ds^2, \quad |h| \ll Ds.$$

The residue inequality $R \ll Ds^2$ is controlled by the 2-adic coordinate. Conditional on the presence of the central pole, let $\ell \geq 0$ be the integer for which this coordinate is divisible by 4^ℓ but not by $4^{\ell+1}$. Then

$$R = 2^{-\ell} R_0, \quad \mathbb{P}(\ell = n) = \frac{3}{4} 4^{-n},$$

where R_0 is a positive random factor whose law does not depend on ℓ . The scale $R \asymp s^2$ therefore has probability of order s^4 . We then prove that the regular value h , with the necessary weights in D and R_0 , has a continuous density which is positive at zero. Hence the restriction $|h| \ll Ds$ contributes one further factor of s . This gives the fifth-order law; the discrete choices of ℓ produce the periodic factor in (1.2).

Notation

Write $e(x) := e^{2\pi i x}$, and let $\mathbf{1}_E$ denote the indicator of E . For a prime p , $v_p(u)$ is the p -adic valuation, with $v_p(0) := \infty$. The symbols $f \ll g$ and $f = O(g)$ mean that $|f|$ is bounded by a constant multiple of g ; subscripts indicate permitted dependence. We write $f \asymp g$ when both inequalities hold.

For $\mathcal{D} := \mathbb{C} \setminus \mathbb{Z}$, $\mathcal{O}(\mathcal{D})$ denotes the holomorphic functions on $\mathcal{D}$, with locally uniform convergence. If $K \subset \mathcal{D}$ is compact, $\|f\|_K := \sup_{z \in K} |f(z)|$. Principal values are symmetric:

$$\text{PV} \sum_{k \in \mathbb{Z}} b_k := \lim_{L \rightarrow \infty} \sum_{|k| \leq L} b_k.$$

2 Preliminaries

2.1 The spectral equation

Translation of the torus moves the scatterer to the origin without changing the spectrum, henceforth we take $o = 0$. The eigenfunctions of $-\Delta$ are $e^{iv \cdot x}$ (we refer to the spectrum of $-\Delta$ without any singularity as the *free spectrum*), with eigenvalues $|v|^2$. On each free eigenspace, evaluation at the origin has a kernel of codimension one. This kernel is the surviving old eigenspace. The new eigenfunctions are multiples of the Green function

$$g_w(x) := \frac{1}{(2\pi)^3} \sum_{v \in \mathbb{Z}^3} \frac{e^{iv \cdot x}}{|v|^2 - w},$$

with convergence in L^2 when w is off the free spectrum. The regularized value at the scatterer gives the scalar spectral equation; see [RU, K] for the self-adjoint extension construction.

Put

$$r(n) := r_3(n), \quad \mathcal{S} := \{n \geq 0 : r(n) > 0\}, \quad \rho := \frac{5}{6}.$$

In particular $r(0) = 1$. For a fixed finite real parameter c , consider

$$G(w) = c, \quad \text{where} \quad G(w) := \sum_{n \geq 0} r(n) \left(\frac{1}{n - w} - \frac{n}{n^2 + 1} \right). \quad (2.1)$$

This is the spectral equation for a fixed nontrivial self-adjoint extension, in the normalization used here; see [K]. For a compact set of w away from the poles and large n ,

$$\left| \frac{1}{n-w} - \frac{n}{n^2+1} \right| \ll_K n^{-2}, \quad \sum_{n \asymp Y} r(n) n^{-2} \ll Y^{-1/2}.$$

The second estimate follows from the elementary lattice bound $\sum_{n \leq Y} r(n) \ll Y^{3/2}$. The dyadic tail is summable, so the series converges locally normally. Differentiation gives

$$G'(x) = \sum_{n \geq 0} \frac{r(n)}{(n-x)^2} > 0.$$

At consecutive poles, G runs from $-\infty$ to $+\infty$, which proves interlacing and simplicity of the new eigenvalues. For $m \in \mathcal{S}$, $m > 0$, write m_- and m_+ for the previous and next elements of $\mathcal{S}$. There is exactly one solution $\lambda_m(c) \in (m_-, m)$. This pole labeling differs from the ordinal labeling $\Lambda_j(c)$ in the introduction: if m_j is the j th positive represented integer, then $\lambda_{m_j}(c) = \Lambda_j(c)$. Set

$$s_m(c) := \rho(\lambda_{m_+}(c) - \lambda_m(c)), \quad N(X) := \#(\mathcal{S} \cap (0, X]).$$

We study only the new eigenvalues, counted once each. In the proof, s_m is indexed by the pole m ; thus s_{m_j} is the j th gap.

2.2 Arithmetic normalization

The three-squares criterion and the disjointness of its exceptional progressions give

$$\begin{aligned} \mathbb{N} \setminus \mathcal{S} &= \bigsqcup_{a \geq 0} \{4^a(8b+7) : b \geq 0\}, \\ X - N(X) &= \frac{X}{8} \sum_{a \geq 0} 4^{-a} + O(\log(2X)) = \frac{X}{6} + O(\log(2X)). \end{aligned}$$

Here the finitely many progressions meeting $[1, X]$ each contribute an $O(1)$ error; the remaining geometric tail contributes $O(1)$. In particular, $N(X) = \rho X + O(\log(2X))$.

The three-squares criterion gives old gaps between 1 and 3. Moreover, $m-1$ and $m+1$ cannot both be omitted, so $m_+ - m_- \leq 4$. Interlacing therefore gives

$$0 < s_m(c) < 4\rho = 10/3. \tag{2.2}$$

Finally, three squares summing to a multiple of four must all be even. Dividing their coordinates by two gives

$$r(4n) = r(n), \quad \frac{r(4n)}{2\pi\sqrt{4n}} = \frac{1}{2} \frac{r(n)}{2\pi\sqrt{n}}.$$

2.3 Congruence probability

The profinite integers form the set of infinite sequences satisfying the usual congruence conditions:

$$\widehat{\mathbb{Z}} := \{(u_M)_{M \geq 1} : u_M \in \mathbb{Z}/M\mathbb{Z}, \quad u_N \equiv u_M \pmod{M} \text{ if } M \mid N\}.$$

By the Chinese remainder theorem

$$\widehat{\mathbb{Z}} \cong \prod_p \mathbb{Z}_p.$$

Here $\mathbb{Z}_p$ is the ring of p -adic integers. Normalized Haar measure is characterized by

$$\mathbb{P}(U \equiv b \pmod{M}) = \frac{1}{M}.$$

Thus the coordinates U_p are independent, and an M -periodic function P satisfies

$$\mathbb{E}P(U) = \frac{1}{M} \sum_{b \bmod M} P(b).$$

The translates $U + k$ use the same coordinates and are dependent. We use $\mathbb{E}$, $\mathbb{P}$, and Law for expectation, probability, and law.

3 Periodic approximation of the multiplicities

For $n \geq 1$ and $M \geq 1$, define

$$a(n) := \frac{r(n)}{2\pi\sqrt{n}}, \quad P_M(b) := \frac{1}{M^2} \#\{v \bmod M : |v|^2 \equiv b \pmod{M}\}.$$

Thus P_M is nonnegative, M -periodic, and has mean one. For a sequence $b(n)$ define the mean-square seminorm (the Besicovitch seminorm) by

$$\|b\|_B^2 := \limsup_{X \rightarrow \infty} \frac{1}{X} \sum_{1 \leq n \leq X} |b(n)|^2.$$

Lemma 3.1. *Let $C := 4\pi^2/(21\zeta(3))$ and $M_Q := \text{lcm}(1, \dots, Q)$. Then*

$$\begin{aligned} \|a - P_M\|_B^2 &= C - \frac{1}{M} \sum_{b \bmod M} P_M(b)^2, \\ \|a - P_{M_Q}\|_B^2 &\ll Q^{-1}. \end{aligned} \tag{3.1}$$

The *limsup* in (3.1) is a limit.

Proof. Set $A_2(x) := \sum_{1 \leq n \leq x} r(n)^2$ and $\kappa := 8\pi^4/(21\zeta(3))$. The input from [CKO] is

$$A_2(x) = \kappa x^2 + o(x^2).$$

Partial summation gives

$$\sum_{1 \leq n \leq X} a(n)^2 = CX + o(X), \quad C = \frac{\kappa}{2\pi^2}. \tag{3.2}$$

Lattice counting in a fixed congruence class gives

$$\#\{x \in v + M\mathbb{Z}^3 : |x| \leq R\} = \frac{4\pi R^3}{3M^3} + O_M(R^2 + 1).$$

Summing over the classes representing b and applying partial summation yields

$$\begin{aligned} \sum_{\substack{0 < n \leq X \\ n \equiv b \pmod{M}}} r(n) &= \frac{4\pi P_M(b)}{3M} X^{3/2} + O_M(X + 1), \\ \sum_{\substack{0 < n \leq X \\ n \equiv b \pmod{M}}} a(n) &= \frac{P_M(b)}{M} X + O_M(\sqrt{X} + 1). \end{aligned}$$

Consequently

$$\lim_{X \rightarrow \infty} \frac{1}{X} \sum_{n \leq X} a(n) P_M(n) = \lim_{X \rightarrow \infty} \frac{1}{X} \sum_{n \leq X} P_M(n)^2 = \frac{1}{M} \sum_{b \bmod M} P_M(b)^2.$$

Expanding the square and using (3.2) proves (3.1), including existence of the limit.

Put $S(q, h) := \sum_{x \bmod q} e(hx^2/q)$. Character orthogonality gives

$$P_M(n) = M^{-3} \sum_{h \bmod M} S(M, h)^3 e(-hn/M).$$

Writing $h/M = h'/q$ in lowest terms and using $S(M, h) = (M/q)S(q, h')$ gives

$$P_M(n) = \sum_{q|M} \sum_{h \bmod q}^* c_{q,h} e(-hn/q), \quad c_{q,h} := q^{-3} S(q, h)^3. \quad (3.3)$$

The star means $(h, q) = 1$, with the constant term included at $q = 1$. Squaring out and again using orthogonality then gives

$$\frac{1}{M} \sum_{b \bmod M} P_M(b)^2 = \sum_{q|M} \sum_{h \bmod q}^* |c_{q,h}|^2.$$

The quadratic Gauss-sum evaluation (see [BEW, pp. 18–28]), for $(h, q) = 1$, is

$$|S(q, h)|^2 = \begin{cases} q, & q \text{ odd}, \\ 0, & q \equiv 2 \pmod{4}, \\ 2q, & 4 \mid q. \end{cases}$$

For each q , this magnitude is independent of the $\varphi(q)$ choices of h coprime to q . Hence the total energy factors as

$$\begin{aligned} \sum_{q \geq 1} \sum_{h \bmod q}^* |c_{q,h}|^2 &= \sum_{\substack{q \geq 1 \\ q \text{ odd}}} \frac{\varphi(q)}{q^3} + 8 \sum_{\substack{q \geq 1 \\ 4 \mid q}} \frac{\varphi(q)}{q^3}, \\ &= \left(1 + \sum_{\nu \geq 2} \frac{8\varphi(2^\nu)}{2^{3\nu}} \right) \sum_{q \text{ odd}} \frac{\varphi(q)}{q^3} \\ &= \frac{4}{3} \cdot \frac{6}{7} \frac{\zeta(2)}{\zeta(3)} = \frac{4\pi^2}{21\zeta(3)} = C. \end{aligned}$$

Here $\sum_{q \geq 1} \varphi(q)q^{-3} = \zeta(2)/\zeta(3)$ and its factor at two is $1 + \sum_{\nu \geq 1} 2^{\nu-1}2^{-3\nu} = 7/6$. Every $q \leq Q$ divides M_Q . Therefore

$$0 \leq \|a - P_{M_Q}\|_B^2 = \sum_{q \nmid M_Q} \sum_h^* |c_{q,h}|^2 \leq 8 \sum_{q > Q} \frac{\varphi(q)}{q^3} \leq 8 \sum_{q > Q} q^{-2} \ll Q^{-1}.$$

□

Remark 3.2. The exact matching of the two second-moment constants is essential. Knowing just an upper bound for $\sum r(n)^2$ would not prove that the periodic approximation error tends to zero.

4 The periodic Green function

The periodic Green function is

$$(\mathcal{H}_M P)(u, z) := -\frac{\pi}{M} \sum_{b=0}^{M-1} P(u+b) \cot\left(\frac{\pi(z-b)}{M}\right), \quad z \in \mathcal{D} := \mathbb{C} \setminus \mathbb{Z}, \quad (4.1)$$

where P is an M -periodic sequence and $u \in \mathbb{Z}/M\mathbb{Z}$. For $P = P_M$, this is the scalar Green function of the periodic arithmetic model. The weights $P_M(u+k)$ retain their arithmetic correlations.

To see where this function comes from, write $\lambda = m + z$ in the spectral equation and use $r_3(n) = 2\pi\sqrt{n} a(n)$ for $n \geq 1$:

$$\frac{G(m+z)}{2\pi\sqrt{m}} = -\frac{1}{2\pi\sqrt{m}(m+z)} + \sum_{n \geq 1} \sqrt{\frac{n}{m}} a(n) \left(\frac{1}{n-m-z} - \frac{n}{n^2+1} \right).$$

Replace a by P_M and set $n = m + k$. For fixed k , the factor $\sqrt{1+k/m}$ tends to one. This suggests the local model

$$\text{PV} \sum_{k \in \mathbb{Z}} \frac{P_M(m+k)}{k-z} = (\mathcal{H}_M P_M)(m, z).$$

The cotangent identity and the passage to this principal value are proved below. In particular, the regularized distant terms leave no additional constant in the limit.

Write $P_Q := P_{M_Q}$ and put

$$H_Q(u, z) := (\mathcal{H}_{M_Q} P_Q)(u, z), \quad F_m^{(c)}(z) := \frac{G(m+z) - c}{2\pi\sqrt{m}}.$$

The main result of this section transfers the mean-square approximation of the multiplicities to the spectral equation.

Lemma 4.1. *For every compact $K \subset \mathcal{D}$ and fixed $c \in \mathbb{R}$,*

$$\limsup_{X \rightarrow \infty} \frac{1}{X} \sum_{X < m \leq 2X} \|F_m^{(c)} - H_Q(m, \cdot)\|_K^2 \leq C_K Q^{-1}.$$

We prove three auxiliary lemmas before returning to the proof of Lemma 4.1. To transfer approximation errors, define the linear operator

$$(T_z b)(m) := \frac{1}{\sqrt{m}} \sum_{n \geq 1} \sqrt{n} b(n) \left(\frac{1}{n-m-z} - \frac{n}{n^2+1} \right).$$

Thus

$$\frac{G(m+z)}{2\pi\sqrt{m}} = (T_z a)(m) - \frac{1}{2\pi\sqrt{m}(m+z)}, \quad T_z a - T_z P_Q = T_z(a - P_Q).$$

For $\|b\|_B < \infty$, the series converges locally normally on $\mathcal{D}$ by Cauchy–Schwarz on dyadic blocks: we have $\sum_{n \leq Y} |b(n)|^2 = O_b(Y)$, and the regularized kernel is $O_{m,K}(n^{-3/2})$.

Lemma 4.2. *For every compact $K \subset \mathcal{D}$, there is a constant C_K , independent of b , such that*

$$\limsup_{X \rightarrow \infty} \frac{1}{X} \sum_{X < m \leq 2X} \sup_{z \in K} |(T_z b)(m)|^2 \leq C_K \|b\|_B^2.$$

Proof. Let $B := \|b\|_B$ and fix $\varepsilon > 0$. There is Y_0 such that

$$\sum_{n \leq Y} |b(n)|^2 \leq (B^2 + \varepsilon)Y \quad (Y \geq Y_0). \quad (4.2)$$

Throughout, $m \in (X, 2X]$, $z \in K$, and X is sufficiently large in terms of K . Split the sum into $n < X/2$, $X/2 \leq n \leq 4X$, and $n > 4X$.

The middle range. Set $f_X(n) := \sqrt{n} b(n) \mathbf{1}_{[X/2, 4X]}(n)$, extended by zero to $\mathbb{Z}$. Writing $k = n - m$, decompose

$$\sum_n \frac{f_X(n)}{n - m - z} = \sum_{k \neq 0} \frac{f_X(m + k)}{k} + \sum_{k \in \mathbb{Z}} r_z(k) f_X(m + k), \quad r_z(k) := \begin{cases} -1/z, & k = 0, \\ z/[k(k - z)], & k \neq 0. \end{cases}$$

Since K is compact and avoids $\mathbb{Z}$, the finitely many small denominators stay away from zero, while $r_z(k) = O_K(k^{-2})$ for large $|k|$. Thus $\sup_{z \in K} \|r_z\|_{\ell^1} < \infty$. The discrete Hilbert transform and Young's inequality give

$$\sum_{m \in \mathbb{Z}} \left| \sum_n \frac{f_X(n)}{n - m - z} \right|^2 \leq C_K \|f_X\|_{\ell^2}^2.$$

Since $m > X$ and $n \leq 4X$ on the support of f_X ,

$$\frac{1}{X} \sum_{X < m \leq 2X} \left| \frac{1}{\sqrt{m}} \sum_n \frac{f_X(n)}{n - m - z} \right|^2 \leq \frac{C_K}{X} \sum_{n \leq 4X} |b(n)|^2 \ll_K B^2 + \varepsilon.$$

The regularizing term is controlled by Cauchy-Schwarz:

$$\left| \frac{1}{\sqrt{m}} \sum_{X/2 \leq n \leq 4X} \frac{n^{3/2} b(n)}{n^2 + 1} \right|^2 \leq \frac{1}{X} \sum_{n \leq 4X} |b(n)|^2 \sum_{X/2 \leq n \leq 4X} \frac{1}{n} \ll B^2 + \varepsilon.$$

The upper tail. Write the regularized kernel as

$$K_{m,z}(n) := \sqrt{\frac{n}{m}} \frac{1 + n(m + z)}{(n - m - z)(n^2 + 1)}.$$

For $n \asymp Y > 4X$ it satisfies $|K_{m,z}(n)| \leq C_K \sqrt{X} Y^{-3/2}$. Hence

$$\begin{aligned} \sum_{n \asymp Y} |K_{m,z}(n) b(n)| &\leq C_K \sqrt{X} Y^{-3/2} \sqrt{Y} \left(\sum_{n \asymp Y} |b(n)|^2 \right)^{1/2} \\ &\leq C_K (B^2 + \varepsilon)^{1/2} (X/Y)^{1/2}. \end{aligned}$$

Summing over $Y = 2^j 4X$ gives $O_K((B^2 + \varepsilon)^{1/2})$. The subtraction must be combined with the resolvent in this range.

The lower tail. On $n \asymp Y < X/2$, its two kernels satisfy

$$\frac{\sqrt{n/m}}{|n - m - z|} \ll_K Y^{1/2} X^{-3/2}, \quad \sqrt{\frac{n}{m}} \frac{n}{n^2 + 1} \ll X^{-1/2} Y^{-1/2}.$$

By (4.2), the corresponding block sums are at most

$$C_K (B^2 + \varepsilon)^{1/2} (Y/X)^{3/2}, \quad C (B^2 + \varepsilon)^{1/2} (Y/X)^{1/2}.$$

Their dyadic sums converge. The fixed set $n < Y_0$ contributes $o_b(1)$. Together these estimates prove

$$\limsup_{X \rightarrow \infty} \frac{1}{X} \sum_{X < m \leq 2X} |T_z b(m)|^2 \leq C_K(B^2 + \varepsilon).$$

To pass to the supremum, cover K by finitely many discs $B(z_j, r_j)$ with $\overline{B(z_j, 2r_j)} \subset \mathcal{D}$. For any holomorphic f on the larger discs, the submean inequality gives

$$\sup_{z \in B(z_j, r_j)} |f(z)|^2 \leq \frac{1}{\pi r_j^2} \int_{B(z_j, 2r_j)} |f(w)|^2 dA(w).$$

Apply the uniform finite- X estimates above under these area integrals, then take $X \rightarrow \infty$ and $\varepsilon \downarrow 0$. $\square$

Lemma 4.3. *For every M -periodic sequence P ,*

$$(\mathcal{H}_M P)(u, z) = \lim_{L \rightarrow \infty} \sum_{|k| \leq L} \frac{P(u+k)}{k-z}.$$

Convergence is locally uniform for $z \in \mathcal{D}$.

Proof. For $k = b + Mj$ and $\zeta := (z - b)/M$, the cotangent partial-fraction expansion [DLMF, Eq. 4.22.3] gives

$$\text{PV} \sum_{j \in \mathbb{Z}} \frac{1}{b + Mj - z} = \frac{1}{M} \text{PV} \sum_{j \in \mathbb{Z}} \frac{1}{j - \zeta} = -\frac{\pi}{M} \cot(\pi \zeta).$$

The cutoff $|b + Mj| \leq L$ differs from a symmetric j -cutoff by at most two terms, each $O_K(L^{-1})$. Summing the M residue classes proves the formula. Pairing positive and negative j leaves an $O_K(j^{-2})$ tail, which proves local uniformity. $\square$

Lemma 4.4. *For every fixed M -periodic sequence P and compact $K \subset \mathcal{D}$,*

$$\sup_{z \in K} |(T_z P)(m) - (\mathcal{H}_M P)(m, z)| \rightarrow 0 \quad (m \rightarrow \infty).$$

In particular there is no additional real constant in this limit.

Proof. Write $w := m + z$. The elementary identity

$$\sqrt{n} \left(\frac{1}{n-w} - \frac{n}{n^2+1} \right) = \frac{w}{\sqrt{n}(n-w)} + \frac{1}{\sqrt{n}(n^2+1)}$$

gives

$$T_z P(m) = \left(1 + \frac{z}{m}\right) \sum_{k \geq 1-m} \frac{P(m+k) \sqrt{m/(m+k)}}{k-z} + O_P(m^{-1/2}). \quad (4.3)$$

Split $P = \bar{P} + P_0$, where P_0 is periodic with mean zero.

Its interval sums satisfy $\sup_{a \leq b} |\sum_{n=a}^b P_0(n)| \leq C_P$. For $L > 2 \sup_K |z| + 2$ and $g_{m,z}(t) := \sqrt{m}/(\sqrt{t}(t-m-z))$,

$$g'_{m,z}(t) = -\frac{\sqrt{m}}{2t^{3/2}(t-m-z)} - \frac{\sqrt{m}}{\sqrt{t}(t-m-z)^2}.$$

Splitting at $m/2$ and $2m$ gives the variation bound

$$|g_{m,z}(1)| + |g_{m,z}(m-L)| + |g_{m,z}(m+L)| + \int_{[1,m-L] \cup [m+L,\infty)} |g'_{m,z}(t)| dt \ll_K m^{-1/2} + L^{-1}.$$

Summation by parts therefore shows that the weighted tails in (4.3) are $O_{P,K}(L^{-1} + m^{-1/2})$. The unweighted periodic tails are $O_{P,K}(L^{-1})$ by the same argument with $(t-m-z)^{-1}$. On $|k| \leq L$,

$$\sqrt{m/(m+k)} = 1 + O(L/m).$$

Take $m \rightarrow \infty$ and then $L \rightarrow \infty$ to obtain the required limit for P_0 .

It remains to treat $P = 1$. On $|k| > L$, replacing $(k-z)^{-1}$ by k^{-1} costs at most

$$C_K \sum_{\substack{n \geq 1 \\ |n-m| > L}} \frac{\sqrt{m/n}}{(n-m)^2} \ll_K L^{-1} + m^{-1}.$$

To control the signed tail, put

$$R_{m,L} := \sum_{\substack{n \geq 1 \\ |n-m| > L}} \frac{\sqrt{m/n}}{n-m}.$$

For fixed $0 < \delta < 1/2$, its portion with $|n/m - 1| \geq \delta$ is a Riemann sum:

$$\sum_{|n/m-1| \geq \delta} \frac{\sqrt{m/n}}{n-m} \rightarrow I_\delta := \int_{(0,1-\delta] \cup [1+\delta,\infty)} \frac{t^{-1/2}}{t-1} dt.$$

The endpoint bounds near zero and infinity are respectively $O(\sqrt{\varepsilon})$ and $O(R^{-1/2})$, uniformly for the sums and integrals. The intervening paired terms satisfy

$$\left| \sum_{L < k < \delta m} \frac{(1+k/m)^{-1/2} - (1-k/m)^{-1/2}}{k} \right| \leq C\delta.$$

Finally,

$$\lim_{\delta \downarrow 0} I_\delta = \text{PV} \int_0^\infty \frac{2 dv}{v^2 - 1} = \left[\log \left| \frac{v-1}{v+1} \right| \right]_0^\infty = 0.$$

Thus $\limsup_m |R_{m,L}| \leq |I_\delta| + C\delta$, so $R_{m,L} \rightarrow 0$ for fixed L . Restoring z and the local terms, then letting $L \rightarrow \infty$, gives

$$T_z 1(m) \rightarrow \text{PV} \sum_{k \in \mathbb{Z}} \frac{1}{k-z} = -\pi \cot(\pi z).$$

All estimates are uniform for $z \in K$; no constant remains. □

Proof of Lemma 4.1. The definitions give the exact identity

$$F_m^{(c)}(z) = T_z a(m) - \frac{1}{2\pi\sqrt{m}(m+z)} - \frac{c}{2\pi\sqrt{m}}.$$

Therefore

$$\begin{aligned} F_m^{(c)}(z) - H_Q(m, z) &= T_z(a - P_Q)(m) + (T_z P_Q(m) - H_Q(m, z)) \\ &\quad - \frac{1}{2\pi\sqrt{m}(m+z)} - \frac{c}{2\pi\sqrt{m}}. \end{aligned}$$

The squared mean of the first term is $O_K(Q^{-1})$ by Lemmas 3.1 and 4.2. For fixed Q , the second term tends uniformly to zero on K by Lemma 4.4. The final terms are $O_K(m^{-3/2})$ and $O_c(m^{-1/2})$. Apply $|x_1 + \dots + x_4|^2 \leq 4 \sum_i |x_i|^2$ and take the limsup. □

5 The limiting meromorphic function

Let $\Omega := \widehat{\mathbb{Z}}$ with normalized Haar measure, and let U be its coordinate random variable. For $M \mid N$, averaging over the N/M lifts of a residue $b \pmod{M}$ gives

$$\begin{aligned}\mathbb{E}[P_N(U) \mid U \equiv b \pmod{M}] &= \frac{M}{N} \frac{1}{N^2} \#\{x \bmod N : |x|^2 \equiv b \pmod{M}\} \\ &= \frac{M}{N^3} \left(\frac{N}{M}\right)^3 M^2 P_M(b) = P_M(b).\end{aligned}$$

Since $M_Q \mid M_{Q+1}$, this is a martingale identity. Its second moments obey

$$\mathbb{E}P_Q(U)^2 \leq C, \quad \mathbb{E}|P_{Q'}(U) - P_Q(U)|^2 = \mathbb{E}P_{Q'}(U)^2 - \mathbb{E}P_Q(U)^2 \quad (Q' \geq Q).$$

The L^2 martingale convergence theorem [D, Theorem 4.4.6, with $p = 2$] therefore gives a nonnegative limit A , with

$$P_Q \longrightarrow A \text{ in } L^2 \text{ and almost surely,} \quad \mathbb{E}A = 1, \quad \mathbb{E}A^2 = C, \quad \mathbb{E}|A - P_Q|^2 \ll Q^{-1}. \quad (5.1)$$

Lemma 5.1. *Up to a Haar null set,*

$$A(U) > 0 \iff U_2 \notin \bigcup_{a \geq 0} 4^a(7 + 8\mathbb{Z}_2).$$

Call this event E and set $I_k(U) := \mathbf{1}_E(U + k)$. Then $\mathbb{P}(E) = 5/6$. Simultaneously for all integer k , $A(U + k) > 0$ if and only if $I_k(U) = 1$, almost surely.

Proof. The issue is strict positivity; nonnegative martingale convergence alone does not settle it. We compute the local factors $\beta_p(u) := \lim_{\nu \rightarrow \infty} P_{p^\nu}(u)$.

For odd p , let

$$N_p(u) := \#\{x \in (\mathbb{Z}/p\mathbb{Z})^3 : |x|^2 \equiv u \pmod{p}\} = p^2 P_p(u).$$

Character orthogonality and the quadratic Gauss sum give

$$N_p(u) = p^2 + p \left(\frac{-u}{p}\right) \quad (u \not\equiv 0 \pmod{p}), \quad N_p(0) = p^2.$$

Every nonzero solution modulo p is nonsingular, and each such solution has $p^{2(\nu-1)}$ lifts to a specified right-hand side modulo p^ν . When $p \mid u$, there are $p^2 - 1$ primitive solutions. The imprimitive solutions have the form $x = py$; if $p^2 \mid u$ their normalized contribution is

$$p^{-2\nu} p^3 \#\{y \bmod p^{\nu-2} : |y|^2 \equiv u/p^2 \pmod{p^{\nu-2}}\} = p^{-1} P_{p^{\nu-2}}(u/p^2).$$

It follows that, for $u \neq 0$,

$$\beta_p(u) = \begin{cases} 1 + p^{-1} \left(\frac{-u}{p}\right), & v_p(u) = 0, \\ 1 - p^{-2}, & v_p(u) = 1, \\ 1 - p^{-2} + p^{-1} \beta_p(u/p^2), & v_p(u) \geq 2. \end{cases}$$

At $u = 0$, the limiting value is $1 + p^{-1}$. The same finite-level recursion, and then bounded convergence, give

$$1 - p^{-1} \leq \beta_p \leq 1 + p^{-1}, \quad \mathbb{E}\beta_p(U_p) = 1, \quad \mathbb{E}(\beta_p(U_p) - 1)^2 \leq p^{-2}.$$

A similar recursive argument at the prime two yields, for $u = 4^a u_0$ with $4 \nmid u_0$,

$$\beta_2(u) = 2^{-a} \begin{cases} 3/2, & u_0 \equiv 1, 2 \pmod{4}, \\ 1, & u_0 \equiv 3 \pmod{8}, \\ 0, & u_0 \equiv 7 \pmod{8}. \end{cases} \quad \beta_2(0) = 0. \quad (5.2)$$

Finite densities are bounded by $3/2$, so $\mathbb{E}\beta_2(U_2) = 1$ and $\mathbb{E}[\beta_p(U_p) \mid U_p \bmod p^\nu] = P_{p^\nu}(U_p)$ for all p .

The odd-prime product $W := \prod_{p>2} \beta_p(U_p)$ converges in L^2 , since

$$\mathbb{E} \prod_{2 < p \leq Y} \beta_p^2 = \prod_{2 < p \leq Y} (1 + O(p^{-2})) \ll 1.$$

For strict positivity, Taylor expansion and independence give

$$\sum_{p>2} |\mathbb{E} \log \beta_p| < \infty, \quad \sum_{p>2} \text{Var}(\log \beta_p) < \infty.$$

The centered logarithmic partial sums are an L^2 -bounded martingale. Their almost-sure convergence shows $0 < W < \infty$ almost surely. By the Chinese remainder theorem,

$$B(U) := \beta_2(U_2)W, \quad \mathbb{E}[B \mid U \bmod M] = P_M(U).$$

The congruence sigma-fields generate the Borel sigma-field of $\widehat{\mathbb{Z}}$; hence $A = B$ in L^2 and almost surely. Formula (5.2) now proves the assertion, apart from the null set $U_2 = 0$. Finally,

$$\mathbb{P}(A = 0) = \sum_{a \geq 0} \frac{1}{8 \cdot 4^a} = \frac{1}{6}.$$

Translation invariance and a countable union of null sets give the simultaneous assertion for every $k \in \mathbb{Z}$. $\square$

Lemma 5.2. *Let $M \geq 1$ and let b be an M -periodic complex sequence. For every compact $K \subset \mathscr{D}$,*

$$\frac{1}{M} \sum_{u \bmod M} \|\mathcal{H}_M b(u, \cdot)\|_K^2 \leq \frac{C_K}{M} \sum_{u \bmod M} |b(u)|^2, \quad (5.3)$$

where C_K is independent of M and b . If $M \mid N$ and b is M -periodic, then $\mathcal{H}_N b = \mathcal{H}_M b$.

Proof. Recall from (4.1) that $\mathcal{H}_M$ acts on M -periodic sequences, and that Lemma 4.3 gives the principal-value representation

$$(\mathcal{H}_M b)(u, z) = \text{PV} \sum_{k \in \mathbb{Z}} \frac{b(u+k)}{k-z}.$$

We first prove an L^2 estimate for this operator. Let

$$\chi_{r,M}(u) := e(ru/M), \quad \widehat{b}(r) := \frac{1}{M} \sum_{u \bmod M} b(u) e(-ru/M), \quad 0 \leq r < M.$$

Thus $b(u) = \sum_{r=0}^{M-1} \widehat{b}(r) \chi_{r,M}(u)$. Applying $\mathcal{H}_M$ to a character and using Lemma 4.3, we find

$$(\mathcal{H}_M \chi_{r,M})(u, z) = \chi_{r,M}(u) \mathfrak{m}_z(r/M), \quad \mathfrak{m}_z(\theta) := \text{PV} \sum_{k \in \mathbb{Z}} \frac{e(k\theta)}{k-z}.$$

Each Fourier character is an eigenfunction of $\mathcal{H}_M$, and the multiplier $\mathfrak{m}_z(\theta)$ is the scalar by which that character is multiplied.

We record the scalar explicitly. For $r = 0$, the cotangent partial-fraction formula gives $\mathfrak{m}_z(0) = -\pi \cot(\pi z)$. If $1 \leq r < M$, then (4.1) gives

$$\mathfrak{m}_z(r/M) = -\frac{\pi}{M} \sum_{a=0}^{M-1} e(ra/M) \cot\left(\frac{\pi(z-a)}{M}\right).$$

To evaluate the finite sum, set $\omega := e(1/M)$ and $\zeta := e(z/M)$. Writing the cotangent in exponential form and summing a geometric progression gives

$$\sum_{a=0}^{M-1} \omega^{ra} \cot\left(\frac{\pi(z-a)}{M}\right) = 2i \sum_{a=0}^{M-1} \frac{\omega^{(r+1)a}}{\zeta - \omega^a} = \frac{2iM\zeta^r}{\zeta^M - 1}.$$

Since $\zeta^M - 1 = 2ie^{\pi iz} \sin(\pi z)$, this proves

$$\mathfrak{m}_z(\theta) = \text{PV} \sum_{k \in \mathbb{Z}} \frac{e(k\theta)}{k - z} = \begin{cases} -\pi \cot(\pi z), & \theta = 0, \\ -\pi e^{i\pi(2\theta-1)z} / \sin(\pi z), & 0 < \theta < 1. \end{cases} \quad (5.4)$$

Here and below θ is represented in $[0, 1)$. If K' is a compact subset of $\mathscr{D}$, then $\sin(\pi z)$ is bounded away from zero on K' , while $|e^{i\pi(2\theta-1)z}| \leq e^{\pi|\Im z|}$. Consequently

$$\sup_{0 \leq \theta < 1} \sup_{z \in K'} |\mathfrak{m}_z(\theta)| < \infty.$$

Fourier inversion and (5.4) now give

$$(\mathcal{H}_M b)(u, z) = \sum_{r=0}^{M-1} \widehat{b}(r) \mathfrak{m}_z(r/M) \chi_{r,M}(u).$$

For each fixed $z \in K'$, Parseval therefore implies

$$\begin{aligned} \frac{1}{M} \sum_{u \bmod M} |(\mathcal{H}_M b)(u, z)|^2 &= \sum_{r=0}^{M-1} |\widehat{b}(r)|^2 |\mathfrak{m}_z(r/M)|^2 \\ &\leq C_{K'} \sum_{r=0}^{M-1} |\widehat{b}(r)|^2 = \frac{C_{K'}}{M} \sum_{u \bmod M} |b(u)|^2. \end{aligned}$$

It remains to make this estimate uniform in $z \in K$. Cover K by finitely many discs $B(z_j, r_j)$ such that $\overline{B(z_j, 2r_j)} \subset \mathscr{D}$, and put $K' := \bigcup_j \overline{B(z_j, 2r_j)}$. If f is holomorphic on the larger discs, the submean inequality gives

$$\sup_{z \in B(z_j, r_j)} |f(z)|^2 \leq \frac{1}{\pi r_j^2} \int_{B(z_j, 2r_j)} |f(w)|^2 dA(w).$$

Apply this to $f(z) = (\mathcal{H}_M b)(u, z)$, sum over j , average over $u \bmod M$, and use the fixed- w Parseval estimate under each area integral. We obtain

$$\frac{1}{M} \sum_{u \bmod M} \|\mathcal{H}_M b(u, \cdot)\|_K^2 \leq \frac{C_K}{M} \sum_{u \bmod M} |b(u)|^2.$$

If $M \mid N$ and b is M -periodic, Lemma 4.3 represents both transforms by the same symmetric principal value. This proves the compatibility assertion. $\square$

Lemma 5.3. *There is a random function $H(U, \cdot) \in \mathcal{O}(\mathcal{D})$ such that, for every compact $K \subset \mathcal{D}$,*

$$\mathbb{E} \|H_Q(U, \cdot) - H(U, \cdot)\|_K^2 \longrightarrow 0. \quad (5.5)$$

Almost surely, H extends meromorphically to $\mathbb{C}$, is real on the real axis away from its poles, and has exactly the poles

$$\{k \in \mathbb{Z} : I_k(U) = 1\}, \quad \text{Res}_{z=k} H(U, z) = -A(U + k).$$

Between consecutive poles it is strictly increasing, with one simple real zero in each such interval.

Proof. If $Q' \geq Q$, then $M_Q \mid M_{Q'}$, and the compatibility in Lemma 5.2 gives

$$H_{Q'}(U, \cdot) - H_Q(U, \cdot) = \mathcal{H}_{M_{Q'}}(P_{Q'} - P_Q)(U, \cdot).$$

The reduction of Haar measure modulo $M_{Q'}$ is uniform, so (5.3) and (5.1) give

$$\mathbb{E} \|H_{Q'} - H_Q\|_K^2 \leq C_K \mathbb{E} |P_{Q'} - P_Q|^2 \leq 2C_K (\mathbb{E} |P_{Q'} - A|^2 + \mathbb{E} |P_Q - A|^2) \ll_K Q^{-1}.$$

Thus (H_Q) is Cauchy in $L^2(\Omega; C(K))$ for every compact $K \subset \mathcal{D}$. Choose a nested compact exhaustion of $\mathcal{D}$ and use the completeness of $C(K)$ on each member of the exhaustion. The resulting compatible limits define a random holomorphic function $H \in \mathcal{O}(\mathcal{D})$ and give (5.5). A diagonal subsequence converges almost surely locally uniformly on $\mathcal{D}$. After intersecting with a further probability-one event, we also have

$$P_Q(U + k) \longrightarrow A(U + k) \quad (k \in \mathbb{Z}),$$

simultaneously for all integers k , by (5.1) and translation invariance.

We work on this probability-one event. Fix $k \in \mathbb{Z}$ and $0 < r < 1/2$. The functions

$$J_{Q,k}(z) := H_Q(U, z) + \frac{P_Q(U + k)}{z - k}$$

are holomorphic on $|z - k| < r$: the second term cancels the only pole of H_Q in that disc. On the boundary circle, local uniform convergence of H_Q and convergence of $P_Q(U + k)$ show that $J_{Q,k}$ converges uniformly. Cauchy's integral formula then gives a holomorphic limit J_k on the disc and uniform convergence on every smaller closed disc. On the punctured disc we therefore have

$$H(U, z) = -\frac{A(U + k)}{z - k} + J_k(z), \quad J_k \text{ holomorphic.}$$

Thus $z = k$ is a pole precisely when $A(U + k) > 0$, and in that case its residue is $-A(U + k)$. Lemma 5.1 identifies these integers with $I_k(U) = 1$. Since $H_Q(U, x)$ are real valued functions for $x \in \mathbb{R} \setminus \mathbb{Z}$, $H(U, x)$ is real too.

For $x \in \mathbb{R} \setminus \mathbb{Z}$ and any surviving pole k ,

$$H'_Q(U, x) = \sum_{j \in \mathbb{Z}} \frac{P_Q(U + j)}{(j - x)^2} \geq \frac{P_Q(U + k)}{(k - x)^2}.$$

The differentiated series is absolutely convergent. Local holomorphic convergence permits differentiation, and passage to the limit gives

$$H'(U, x) \geq \frac{A(U + k)}{(k - x)^2} > 0.$$

By continuity the derivative remains positive at every removable integer point. If $k < l$ are consecutive poles, the residue description gives

$$H(x) \longrightarrow -\infty \ (x \downarrow k), \quad H(x) \longrightarrow +\infty \ (x \uparrow l), \quad H'(x) > 0.$$

The intermediate value theorem proves existence and uniqueness of the zero, and the derivative inequality proves its simplicity. $\square$

6 Convergence of the spacing measures

On the event E , let

$$k_- := \max\{k < 0 : I_k(U) = 1\}, \quad k_+ := \min\{k > 0 : I_k(U) = 1\}.$$

The residue-class argument proving (2.2) applies equally over $\mathbb{Z}_2$: $k_- \geq -3$, $k_+ \leq 3$ and $k_+ - k_- \leq 4$. Let $Z_-(U)$ and $Z_+(U)$ be the unique zeros of $H(U, \cdot)$ in $(k_-, 0)$ and $(0, k_+)$. Define

$$\mu := \text{Law}(\rho(Z_+(U) - Z_-(U)) \mid E). \quad (6.1)$$

This characterizes the limiting law entirely in terms of congruence counts, cotangents, and Haar measure.

Lemma 6.1. *If m is uniformly sampled from the integers in $(X, 2X]$, then*

$$\left(F_m^{(c)}, (\mathbf{1}_{\mathcal{S}}(m+k))_{-3 \leq k \leq 3}\right) \implies \left(H(U, \cdot), (I_k(U))_{-3 \leq k \leq 3}\right) \quad (6.2)$$

in $\mathcal{O}(\mathcal{D})$ with the compact-open topology, times a finite discrete space.

Proof. Equip $\mathcal{O}(\mathcal{D})$ with the bounded compact-open metric

$$d(f, g) := \sum_{j \geq 1} 2^{-j} \min(1, \|f - g\|_{K_j}),$$

where (K_j) is a cofinal compact exhaustion. Truncating this series and applying Lemma 4.1 gives

$$\lim_{Q \rightarrow \infty} \limsup_{X \rightarrow \infty} \mathbb{E}_X d(F_m^{(c)}, H_Q(m, \cdot)) = 0. \quad (6.3)$$

Here $\mathbb{E}_X$ denotes uniform integer sampling in $(X, 2X]$.

For the pole indicators, put

$$J_R(u) := 1 - \sum_{\ell=0}^{R-1} \mathbf{1}_{4^\ell(7+8\mathbb{Z}_2)}(u), \quad L_R := 2^{2R+1}.$$

The function J_R is L_R -periodic. The omitted exceptional classes have Haar mass $4^{-R}/6$; counting these same progressions in an interval gives, uniformly for $|k| \leq 3$,

$$\begin{aligned} \mathbb{P}(J_R(U+k) \neq I_k(U)) &= 4^{-R}/6, \\ \mathbb{P}_X(J_R(m+k) \neq \mathbf{1}_{\mathcal{S}}(m+k)) &\ll 4^{-R} + X^{-1} \log(2X). \end{aligned}$$

The second estimate uses only $O(\log X)$ progressions, each contributing its expected count with an $O(1)$ error.

Choose $R(Q) \rightarrow \infty$ with $L_{R(Q)} \mid M_Q$. On the Polish space

$$\mathcal{X} := \mathcal{O}(\mathcal{D}) \times \{0, 1\}^7, \quad \Delta((f, i), (g, j)) := d(f, g) + \mathbf{1}_{\{i \neq j\}},$$

define

$$\begin{aligned} Y_X(m) &:= \left(F_m^{(c)}, (\mathbf{1}_{\mathcal{S}}(m+k))_{|k| \leq 3} \right), \\ Y_Q(u) &:= \left(H_Q(u, \cdot), (J_{R(Q)}(u+k))_{|k| \leq 3} \right), \\ Y(U) &:= \left(H(U, \cdot), (I_k(U))_{|k| \leq 3} \right). \end{aligned}$$

Equation (6.3), the indicator estimates, and Lemma 5.3 imply

$$\lim_{Q \rightarrow \infty} \limsup_{X \rightarrow \infty} \mathbb{E}_X \Delta(Y_X, Y_Q(m)) = 0, \quad \mathbb{E} \Delta(Y_Q(U), Y(U)) \rightarrow 0.$$

For fixed Q , equidistribution of integers modulo M_Q gives $Y_Q(m) \Rightarrow Y_Q(U)$ as $X \rightarrow \infty$. Testing against bounded Lipschitz functions and using the two approximation bounds proves (6.2). $\square$

Lemma 6.2. *Suppose meromorphic functions f_j with poles only at integers converge locally uniformly on $\mathcal{D}$ to f . Suppose each function is real and strictly increasing between its actual poles, with one zero in each interval between consecutive poles. Suppose also that the pole indicators on $[-3, 3] \cap \mathbb{Z}$ eventually agree with those of f , that 0 is a pole of f , and that the neighboring poles of f lie in $[-3, 3]$. If all actual residues of f are strictly negative, the zeros of f_j immediately to the left and right of 0 converge to the corresponding zeros of f .*

Proof. The neighboring poles eventually equal fixed integers $k_- < 0 < k_+$. Any integer inside $(k_-, 0) \cup (0, k_+)$ is eventually removable for every f_j . For a small disc about such an integer, uniform convergence on its boundary and the maximum principle give uniform convergence throughout the disc. Thus $f_j \rightarrow f$ locally uniformly on both real intervals, including their removable integer points.

Let z_- be the zero in $(k_-, 0)$. For every sufficiently small $\varepsilon > 0$,

$$f(z_- - \varepsilon) < 0 < f(z_- + \varepsilon).$$

Uniform convergence at these two points gives the same signs for f_j when j is large. Its unique zero $z_{-,j}$ must therefore satisfy

$$z_- - \varepsilon < z_{-,j} < z_- + \varepsilon.$$

Letting $\varepsilon \downarrow 0$ proves $z_{-,j} \rightarrow z_-$. The identical argument on $(0, k_+)$ gives $z_{+,j} \rightarrow z_+$. $\square$

Proof of Theorem 1.1. Fix $f \in C([0, 10/3])$. We first average over a dyadic interval of integers, without conditioning the central integer to lie in $\mathcal{S}$. Take any sequence of positive integers $X_n \rightarrow \infty$. By Lemma 6.1 and the Skorokhod representation theorem [Bi, Theorem 6.7], copies

$$Y_n = (F_n, (\varepsilon_{n,k})_{|k| \leq 3}), \quad Y = (H, (I_k(U))_{|k| \leq 3})$$

may be realized on one probability space so that $Y_n \rightarrow Y$ almost surely. For each n with $\varepsilon_{n,0} = 1$, let $Z_{n,-}$ and $Z_{n,+}$ be the zeros of F_n adjacent to the origin; when $\varepsilon_{n,0} = 0$, set $Z_{n,-} = Z_{n,+} = 0$. Likewise set $Z_-(U) = Z_+(U) = 0$ off E . These variables are measurable: on an interval (a, b) between marked poles, the unique zero of an increasing function g is

$$\inf\{r \in (\mathbb{Q} \setminus \mathbb{Z}) \cap (a, b) : g(r) > 0\}.$$

The discrete coordinates are eventually equal. On $I_0(U) = 1$, Lemma 6.2 gives

$$Z_{n,-} \longrightarrow Z_-(U), \quad Z_{n,+} \longrightarrow Z_+(U),$$

while on $I_0(U) = 0$ all four quantities are eventually zero. Hence

$$\varepsilon_{n,0} f(\rho(Z_{n,+} - Z_{n,-})) \longrightarrow I_0(U) f(\rho(Z_+(U) - Z_-(U)))$$

almost surely. The variables are bounded by $\|f\|_\infty$, so their expectations converge. Since the coupled variables retain their marginal laws, these expectations are

$$\lim_{n \rightarrow \infty} \frac{1}{X_n} \sum_{\substack{X_n < m \leq 2X_n \\ m \in \mathcal{S}}} f(s_m(c)) = \mathbb{E} [\mathbf{1}_E(U) f(\rho(Z_+(U) - Z_-(U)))] = \rho \int f d\mu.$$

The last equality uses $\mathbb{P}(E) = \rho$ and (6.1). The sequence (X_n) was arbitrary, and changing an endpoint by $O(1)$ contributes $o(1)$. Hence

$$\frac{1}{X} \sum_{\substack{X < m \leq 2X \\ m \in \mathcal{S}}} f(s_m(c)) \longrightarrow \rho \int f d\mu \quad (f \in C([0, 10/3])).$$

Summing dyadically and letting $N(X)/X \rightarrow \rho$ then gives

$$\frac{1}{N(X)} \sum_{\substack{0 < m \leq X \\ m \in \mathcal{S}}} f(s_m(c)) \longrightarrow \int f d\mu. \quad (6.4)$$

Taking $X = m_N$ in (6.4) then gives

$$\int f d\mu_N^{(c)} = \frac{1}{N} \sum_{j=1}^N f(s_j(c)) \longrightarrow \int f d\mu,$$

which proves (1.1). The limiting function in Lemma 6.1 is the same for every fixed c : in the normalized spectral equation the only c -dependent term is $-c/(2\pi\sqrt{m})$, which tends to zero. Thus the measure μ is independent of the interaction parameter.

The remaining properties of μ follow more or less directly from the preceding work. $\square$

7 Level repulsion

Section 6 identifies the limiting measure with the two zeros of $H(U, \cdot)$ adjacent to a pole at the origin. On the event $E := \{A(U) > 0\}$, write these zeros as $-u(U) < 0 < v(U)$ and set $d(U) := u(U) + v(U)$. Before unfolding, the small-gap distribution is

$$F_{\text{raw}}(s) := \mathbb{P}(d(U) \leq s \mid E), \quad \mu([0, s]) = F_{\text{raw}}(s/\rho).$$

We shall determine the asymptotic behavior of $F_{\text{raw}}(s)$ as $s \downarrow 0$.

We first derive Theorem 1.2 from the required density estimate and then prove that estimate. Its strict positivity is proved in Section 8.

7.1 The local Green function and small gaps

The factorization $\widehat{\mathbb{Z}} = \mathbb{Z}_2 \times \prod_{p>2} \mathbb{Z}_p$ separates the two parts of the argument. Write $U = (t, V)$, where $t \in \mathbb{Z}_2$ and $V = (V_p)_{p>2}$ is Haar on

$$\Omega_o := \prod_{p>2} \mathbb{Z}_p.$$

By Lemma 5.1, the limiting residue field factors as

$$A(U + k) = \beta_2(t + k) \prod_{p>2} \beta_p(V_p + k).$$

The odd-prime product is positive almost surely. Thus the event E is exactly $\{\beta_2(t) > 0\}$, while the size of the central residue is governed by the 2-adic coordinate. To keep the odd-prime randomness visible, put

$$W(V) := \prod_{p>2} \beta_p(V_p), \quad A_t(k) := \beta_2(t + k)W(V + k), \quad R_t := A_t(0).$$

Here R_t is the residue size at the central pole.

We next define the two coefficients that remain after the pole at the origin is removed:

$$h_t := \text{PV} \sum_{k \neq 0} \frac{A_t(k)}{k}, \quad D_t := \sum_{k \neq 0} \frac{A_t(k)}{k^2}. \quad (7.1)$$

The principal value is the symmetric limit in $L^2(\Omega_o)$, and $D_t > 0$ almost surely. Separating the central pole gives

$$H_t(z) := -\frac{R_t}{z} + h_t + \sum_{k \neq 0} A_t(k) \left(\frac{1}{k - z} - \frac{1}{k} \right). \quad (7.2)$$

For Haar-almost every (t, V) this is precisely the function $H(U, z)$ constructed in Section 5. Indeed, the identity holds for the periodic approximants; (5.1) and the operator bounds in Lemma 7.4 permit passage to the limit. Normal convergence away from $\mathbb{Z}$ follows from the summable regularized kernel.

The term $-R_t/z$ is the singular part of H_t at the origin. After removing it, the function

$$F_t(z) := H_t(z) + \frac{R_t}{z}$$

extends analytically across $z = 0$. We call F_t the regular part of H_t at the origin. In particular,

$$F_t(0) = h_t, \quad F'_t(0) = D_t.$$

Positivity of the coefficients gives, uniformly for $|x| \leq 1/2$,

$$F'_t(x) = D_t(1 + O(|x|)), \quad F_t(x) = h_t + D_t x + O(D_t x^2).$$

Thus R_t, h_t, D_t are not auxiliary parameters: they are exactly the pole strength, the regular value, and the regular slope that determine the two nearby zeros to second order.

On E , let $-u_t < 0 < v_t$ be the adjacent zeros of H_t and put $d_t := u_t + v_t$. The following elementary calculation turns the small-gap problem into an inequality involving these three local coefficients.

Lemma 7.1. For $R_t > 0$ set

$$q_t := \frac{\sqrt{h_t^2 + 4D_t R_t}}{D_t}.$$

There exist $\varepsilon_0, C > 0$ such that

$$q_t \leq \varepsilon_0 \implies |d_t - q_t| \leq Cq_t^2,$$

and

$$d_t \leq \varepsilon_0 \implies |q_t - d_t| \leq Cd_t^2.$$

Proof. Suppress the subscript t and write $R, h, D, H, F, u, v, d, q$ for the quantities above. Thus

$$H(x) = -\frac{R}{x} + F(x), \quad F(x) = h + Dx + O(Dx^2), \quad F'(x) = D(1 + O(|x|)).$$

We first assume that q is small. Replace $F(x)$ by its linear approximation $h + Dx$ and consider

$$\tilde{H}(x) := -\frac{R}{x} + h + Dx.$$

Its two roots are

$$x_{\pm} = \frac{-h \pm \sqrt{h^2 + 4DR}}{2D}, \quad x_- < 0 < x_+, \quad x_+ - x_- = q.$$

Because the roots have opposite signs, $|x_{\pm}| \leq q$. Moreover,

$$H(x) = \tilde{H}(x) + O(Dx^2),$$

so that

$$H(x_{\pm}) = O(Dx_{\pm}^2).$$

For $|x|$ sufficiently small,

$$H'(x) = \frac{R}{x^2} + F'(x) \geq \frac{D}{2}.$$

Moving a distance $C_1 x_{\pm}^2$ away from $x_{\pm}$ therefore changes H by at least $(D/2)C_1 x_{\pm}^2$. If C_1 is sufficiently large, this change dominates the error $H(x_{\pm}) = O(Dx_{\pm}^2)$. Hence H has opposite signs at the two endpoints $x_{\pm} - C_1 x_{\pm}^2$ and $x_{\pm} + C_1 x_{\pm}^2$. The true zeros $z_- = -u$ and $z_+ = v$ consequently satisfy

$$z_{\pm} = x_{\pm} + O(x_{\pm}^2).$$

Therefore

$$d = z_+ - z_- = (x_+ - x_-) + O(x_+^2 + x_-^2) = q + O(q^2).$$

We now prove the converse. Assume that $d = u + v$ is small. Since $-u$ and v are zeros of H , we have

$$F(v) = \frac{R}{v}, \quad F(-u) = -\frac{R}{u}.$$

Subtracting these identities gives

$$F(v) - F(-u) = R \left(\frac{1}{v} + \frac{1}{u} \right) = \frac{Rd}{uv}.$$

On the other hand, the derivative estimate for F gives

$$F(v) - F(-u) = \int_{-u}^v F'(x) dx = Dd + O(Dd^2).$$

Cancelling d from the last two displays yields

$$R = Duv(1 + O(d)).$$

Next, the identity $F(v) = R/v$ and the Taylor expansion of F give

$$h + Dv + O(Dv^2) = \frac{R}{v} = Du(1 + O(d)).$$

Since $u, v \leq d$, this implies

$$h = D(u - v) + O(Dd^2).$$

Consequently,

$$\begin{aligned} h^2 + 4DR &= D^2(u - v)^2 + 4D^2uv + O(D^2d^3) \\ &= D^2(u + v)^2 + O(D^2d^3) \\ &= D^2d^2(1 + O(d)). \end{aligned}$$

Here we used the elementary identity $(u - v)^2 + 4uv = (u + v)^2$. Taking the positive square root and dividing by D now gives

$$q = d\sqrt{1 + O(d)} = d + O(d^2),$$

as required. □

For $s > 0$, define

$$\mathcal{Q}_t(s) := \{q_t \leq s\}.$$

Since $D_t > 0$, the definition of q_t gives the exact identity

$$\mathcal{Q}_t(s) = \left\{ R_t \leq \frac{s^2 D_t}{4}, \quad |h_t| \leq (s^2 D_t^2 - 4D_t R_t)^{1/2} \right\}. \quad (7.3)$$

In particular,

$$\mathcal{Q}_t(s) \subseteq \left\{ R_t \leq \frac{s^2 D_t}{4} \right\} \cap \{|h_t| \leq s D_t\}.$$

Thus the residue R_t must be of order at most $s^2 D_t$, and the regular value h_t must lie in an interval of length at most $2s D_t$ around the origin.

We now rewrite the residue inequality in terms of the 2-adic coordinate. Conditional on $\beta_2(t) > 0$, there are unique $\ell \geq 0$ and $b \in \{1, 3/2\}$ such that

$$t = 4^\ell t_0, \quad 4 \nmid t_0, \quad b := \beta_2(t_0).$$

The explicit formula (5.2) gives

$$\mathbb{P}(\ell = n) = \frac{3}{4} 4^{-n} \quad (n \geq 0), \quad \mathbb{P}(b = 1) = \frac{1}{5}, \quad \mathbb{P}(b = 3/2) = \frac{4}{5}. \quad (7.4)$$

The variables ℓ, b, V are independent, and

$$R_t = 2^{-\ell} b W. \quad (7.5)$$

Substitution into (7.3) yields

$$\mathcal{Q}_t(s) = \left\{ 2^{-\ell} bW \leq \frac{s^2 D_t}{4}, \quad |h_t| \leq (s^2 D_t^2 - 4D_t 2^{-\ell} bW)^{1/2} \right\}.$$

For an integer n satisfying $2^{-n} \asymp s^2$, we have

$$\mathbb{P}(\ell = n) = \frac{3}{4} 4^{-n} \asymp s^4.$$

For each such index, the permitted interval for h_t has length

$$2(s^2 D_t^2 - 4D_t 2^{-\ell} bW)^{1/2} \leq 2s D_t.$$

Corollary 7.3 controls this interval jointly with D_t and W and yields the remaining factor s . More precisely, if

$$\log_2(s^{-2}) = L + \alpha, \quad L \in \mathbb{Z}, \quad 0 \leq \alpha < 1,$$

and $\ell = L + j$, then

$$2^{-\ell} = s^2 2^{\alpha-j}, \quad 4^{-\ell} = s^4 2^{2(\alpha-j)}.$$

The dependence on the fractional part α produces the periodic factor in the final asymptotic.

7.2 From smoothing to the small-gap law

To evaluate (7.3), we need the distribution of h_t in an interval of length s , jointly with the variables D_t and W . Moreover, on the relevant classes $\ell \rightarrow \infty$, so $t \rightarrow 0$ in $\mathbb{Z}_2$. The needed input is therefore a density in the h_t variable that is continuous as $(t, h_t) \rightarrow (0, 0)$, with enough uniform moment control in D_t and W to sum over all dyadic classes.

Put

$$\omega(D, W) := 1 + D + D^{-1} + W + W^{-1}, \quad (D, W) \in (0, \infty)^2.$$

Call ψ admissible if $\psi \in C^2((0, \infty)^2)$ and, for some finite C, m ,

$$|\partial^\gamma \psi(D, W)| \leq C \omega(D, W)^m \quad \text{for every } |\gamma| \leq 2.$$

This includes smooth compactly supported tests, constants, and powers of D, W and their reciprocals.

Proposition 7.2. *For every admissible ψ ,*

$$\sup_{t \in \mathbb{Z}_2} \left| \mathbb{E}_V[\psi(D_t, W) e^{i\xi h_t}] \right| \ll_\psi (1 + |\xi|)^{-3/2}. \quad (7.6)$$

The role of the weight ψ is to retain the dependence of h_t on D_t and W ; an unweighted density for h_t would not control the region in (7.3). Only integrability of the Fourier bound is needed. We first use the proposition to finish the small-gap calculation. Its proof occupies Sections 7.3–7.5.

Corollary 7.3. *There are finite positive measures $\nu_{t,y}$ on $(0, \infty)^2$ such that*

$$\mathbb{P}(h_t \in dy, (D_t, W) \in dD dW) = dy \nu_{t,y}(dD, dW). \quad (7.7)$$

They are weakly continuous in $(t, y) \in \mathbb{Z}_2 \times \mathbb{R}$, and for every finite $m \geq 0$,

$$\sup_{t,y} \int \omega(D, W)^m \nu_{t,y}(dD, dW) < \infty. \quad (7.8)$$

For each admissible ψ , the density $f_\psi(t, y) := \int \psi d\nu_{t,y}$ is continuous.

Proof. Write $Z_t := (D_t, W)$. For admissible ψ , the finite signed measure

$$\lambda_{t,\psi}(B) := \mathbb{E}_V [\psi(Z_t) \mathbf{1}_{\{h_t \in B\}}]$$

has integrable Fourier transform by (7.6). Fourier inversion [D, Theorem 3.3.14] therefore gives the continuous density

$$f_\psi(t, y) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{-i\xi y} \mathbb{E}_V [\psi(Z_t) e^{i\xi h_t}] d\xi.$$

Lemma 7.5, (7.16), and truncation in ω show that the integrand is continuous in t for each ξ ; the majorant (7.6) makes f_ψ jointly continuous in (t, y) .

If $\psi \geq 0$, then $\lambda_{t,\psi}$ is positive, so its continuous density is nonnegative. Thus, for fixed (t, y) , the map $\psi \mapsto f_\psi(t, y)$ is a positive functional on $C_c^\infty((0, \infty)^2)$, bounded by $\|\psi\|_\infty f_1(t, y)$. It extends to $C_c((0, \infty)^2)$, and Riesz representation gives a finite positive measure $\nu_{t,y}$.

Choose smooth cutoffs $0 \leq \chi_R \leq 1$ with $\chi_R \uparrow 1$, equal to one on $\{\omega \leq R\}$ and supported on $\{\omega \leq 2R\}$. Positivity of weighted densities gives, for every $m > 0$,

$$0 \leq f_1(t, y) - f_{\chi_R}(t, y) = f_{1-\chi_R}(t, y) \leq R^{-m} f_{\omega^m}(t, y) \ll_m R^{-m}.$$

Hence $\nu_{t,y}((0, \infty)^2) = f_1(t, y)$. More generally, for $r > 0$,

$$0 \leq f_{\omega^m}(t, y) - f_{\omega^m \chi_R}(t, y) \leq R^{-r} f_{\omega^{m+r}}(t, y).$$

Letting $R \rightarrow \infty$ gives

$$\int \omega^m d\nu_{t,y} = f_{\omega^m}(t, y), \quad \sup_{t,y} f_{\omega^m}(t, y) < \infty,$$

which proves (7.8); the same cutoff argument extends $\int \psi d\nu_{t,y} = f_\psi(t, y)$ to every admissible ψ . Continuity on compactly supported tests, together with (7.8), gives weak continuity of $(t, y) \mapsto \nu_{t,y}$. Finally, for $g \in C_c(\mathbb{R})$ and $\psi \in C_c((0, \infty)^2)$,

$$\int_{\mathbb{R}} g(y) \int \psi d\nu_{t,y} dy = \mathbb{E}_V [g(h_t) \psi(Z_t)],$$

and a monotone-class argument gives (7.7). □

The measure relevant to a small gap is therefore

$$\nu_* := \nu_{0,0} = \lim_{(t,y) \rightarrow (0,0)} \nu_{t,y}.$$

It is finite and has moments of every order appearing below. Section 8 proves that its total mass is positive.

We can now state the coefficient in the small-gap asymptotic. Put $\pi_1 := 1/5$, $\pi_{3/2} := 4/5$. For $\alpha \in \mathbb{R}$, set $\eta_j := 2^{\alpha-j}$ and define

$$\Phi(\alpha) := \frac{3}{2} \sum_{j \in \mathbb{Z}} \eta_j^2 \sum_{b \in \{1, 3/2\}} \pi_b \int (D^2 - 4DbW \eta_j)_+^{1/2} \nu_*(dD, dW). \quad (7.9)$$

We prove

$$F_{\text{raw}}(s) = s^5 \Phi(\log_2(s^{-2})) + o(s^5). \quad (7.10)$$

The coefficient in Theorem 1.2 is then

$$\Psi(\alpha) := \rho^{-5} \Phi(\alpha + 2 \log_2 \rho). \quad (7.11)$$

Proof of Theorem 1.2. Let

$$Q(s) := \mathbb{P}(q_t \leq s \mid \beta_2(t) > 0), \quad \log_2(s^{-2}) = L + \alpha,$$

where $L \in \mathbb{Z}$ and $0 \leq \alpha < 1$, and put $\eta_j := 2^{\alpha-j}$. For $\ell \geq 0$ and $b \in \{1, 3/2\}$ write

$$\bar{\nu}_{\ell,b,y} := \mathbb{E}_{t|\ell,b}[\nu_{t,y}].$$

Since $2^{-\ell} = s^2 \eta_j$ and $4^{-\ell} = s^4 \eta_j^2$ when $\ell = L + j$, equations (7.4), (7.5), (7.3), and (7.7) give the single identity

$$\frac{Q(s)}{s^5} = \frac{3}{4} \sum_{b \in \{1, 3/2\}} \pi_b \sum_{\substack{j \in \mathbb{Z} \\ j \geq -L}} \eta_j^2 \int_{\mathbb{R}} \int \mathbf{1}_{\{z^2 \leq D^2 - 4DbW\eta_j\}} \bar{\nu}_{L+j,b,sz}(dD, dW) dz. \quad (7.12)$$

The indicator forces $D \geq |z|$ and $D/W \geq 4b\eta_j$. Since $(D/W)^3 \max(1, D^2) \ll \omega(D, W)^8$, (7.8) gives, uniformly in every parameter,

$$\int \mathbf{1}_{\{z^2 \leq D^2 - 4DbW\eta_j\}} \bar{\nu}_{L+j,b,sz}(dD, dW) \ll (1 + z^2)^{-1} \min(1, \eta_j^{-3}). \quad (7.13)$$

Consequently

$$\sum_{j \in \mathbb{Z}} \eta_j^2 \min(1, \eta_j^{-3}) \ll 1 \quad (0 \leq \alpha < 1),$$

so (7.12) has an integrable majorant on $\mathbb{Z} \times \mathbb{R}$ and, in particular, $Q(s) \ll s^5$.

Now let $s_n \downarrow 0$ with $\alpha(s_n) \rightarrow \alpha_0$. For fixed j , $L_n + j \rightarrow \infty$, and every t in the corresponding class lies in $4^{L_n+j}\mathbb{Z}_2$. Corollary 7.3 therefore gives

$$\bar{\nu}_{L_n+j,b,s_n z} \implies \nu_*, \quad \eta_j(s_n) \longrightarrow \eta_j^0 := 2^{\alpha_0-j}.$$

For each (D, W) , the boundary equation $z^2 = D^2 - 4DbW\eta_j^0$ has at most two solutions in z ; Fubini shows that the limiting boundary has ν_* -mass zero for almost every z . After restricting to $|z| \leq R$ and $\omega(D, W) \leq R$, the defining inequalities converge uniformly. Weak convergence and the boundary statement therefore give convergence of the corresponding integrals; (7.13) and (7.8) remove the truncation uniformly. Dominated convergence in (7.12) now gives

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{Q(s_n)}{s_n^5} &= \frac{3}{4} \sum_{j \in \mathbb{Z}} (\eta_j^0)^2 \sum_b \pi_b \int_{\mathbb{R}} \int \mathbf{1}_{\{z^2 \leq D^2 - 4DbW\eta_j^0\}} d\nu_* dz \\ &= \frac{3}{2} \sum_{j \in \mathbb{Z}} (\eta_j^0)^2 \sum_b \pi_b \int (D^2 - 4DbW\eta_j^0)_+^{1/2} d\nu_* = \Phi(\alpha_0). \end{aligned}$$

The same majorant proves continuity of Φ , while $\eta_j(\alpha + 1) = \eta_{j-1}(\alpha)$ proves one-periodicity. Compactness of $[0, 1]$ and the preceding subsequence argument give

$$Q(s) = s^5 \Phi(\log_2(s^{-2})) + o(s^5).$$

Lemma 7.1 gives

$$Q(s - Cs^2) \leq F_{\text{raw}}(s) \leq Q(s + Cs^2).$$

Continuity and periodicity of Φ therefore prove (7.10). Proposition 8.1 gives $\nu_* \neq 0$; on a compact box of positive ν_* -mass the integrand in (7.9) is positive for all sufficiently small η_j . Thus $\Phi > 0$, and continuity and periodicity give $\min \Phi > 0$. Unfolding gives (7.11) and (1.2). Finally Theorem 1.1 and Portmanteau [Bi, Theorem 2.1] give, for each fixed $s > 0$,

$$\limsup_{N \rightarrow \infty} \frac{1}{N} \#\{j \leq N : s_j(c) \leq s\} \leq \mu([0, s]) \ll s^5.$$

□

7.3 Uniform estimates for the local variables

It remains to prove Proposition 7.2. This subsection establishes two preliminary facts. The local products and the Green coefficients have moments of every fixed order, uniformly in the 2-adic parameter t . They also vary continuously with t . We begin by recording the Fourier information on the local densities that will later produce cancellation.

The Gauss-sum computation used in (3.3) gives, for p odd, $\nu \geq 1$, and $(a, p) = 1$,

$$|\widehat{\beta}_p(a/p^\nu)| = p^{-3\nu/2}.$$

The constant Fourier coefficient is one. For $\delta_p := \beta_p - 1$ this yields

$$\mathbb{E}\delta_p = 0, \quad |\delta_p| \leq p^{-1}, \quad v_p := \mathbb{E}\delta_p^2 = \sum_{\nu \geq 1} \varphi(p^\nu) p^{-3\nu} = \frac{1}{p(p+1)}.$$

If $p \nmid k - l$, the covariance is exactly

$$\mathbb{E}[\delta_p(V_p + k)\delta_p(V_p + l)] = -p^{-3}. \quad (7.14)$$

Indeed, if

$$c_q(r) := \sum_{\substack{a \bmod q \\ (a, q) = 1}} e(ar/q),$$

then Parseval expresses the covariance as $\sum_{\nu \geq 1} p^{-3\nu} c_{p^\nu}(k - l)$. When $p \nmid k - l$, $c_p(k - l) = -1$ and the terms with $\nu \geq 2$ vanish.

At the prime two, write

$$\beta_2(t) = \sum_{\alpha \in \mathbb{Z}[1/2]/\mathbb{Z}} c_2(\alpha) e_\alpha(t).$$

For nonzero α , let $q(\alpha)$ denote its denominator in lowest terms. The constant coefficient is one, the coefficients with $q(\alpha) = 2$ vanish, and every coefficient with $q(\alpha) = 2^\nu$, $\nu \geq 2$, has modulus $2^{3/2} 2^{-3\nu/2}$. Consequently

$$\sum_{\alpha} |c_2(\alpha)| = 2 + \sqrt{2}, \quad \sum_{q(\alpha) > 2^{2\ell}} |c_2(\alpha)| \ll 2^{-\ell}. \quad (7.15)$$

This series converges uniformly, including at zero.

For a function C on Ω_o put

$$\mathcal{H}_t C := \text{PV} \sum_{k \neq 0} \frac{\beta_2(t + k) C(V + k)}{k}, \quad \mathcal{D}_t C := \sum_{k \neq 0} \frac{\beta_2(t + k) C(V + k)}{k^2}.$$

Lemma 7.4. *For $1 < q < \infty$,*

$$\|\mathcal{H}_t C\|_q + \|\mathcal{D}_t C\|_q \leq C_q \|C\|_q,$$

uniformly in t . For $\ell \geq 0$, if $t - t' \in 4^\ell \mathbb{Z}_2$, then

$$\|\mathcal{H}_t - \mathcal{H}_{t'}\|_{q \rightarrow q} + \|\mathcal{D}_t - \mathcal{D}_{t'}\|_{q \rightarrow q} \ll_q 2^{-\ell}.$$

Proof. The discrete Hilbert transform is bounded on $\ell^q(\mathbb{Z})$ [BK, Theorem 1.1]. Modulation and Poisson averaging give the same uniform bound for the kernels $r^{|k|}e(k\alpha)/k$, $0 < r < 1$. Calderón's transference principle [Ca], applied to the shift $V \mapsto V + 1$, therefore gives

$$\left\| \sum_{k \neq 0} \frac{r^{|k|}e(k\alpha)C(V+k)}{k} \right\|_q \leq C_q \|C\|_q.$$

As $r \uparrow 1$, density of character polynomials gives a strong L^q limit. In L^2 , the uniformly bounded symmetric partial-sum multipliers identify this limit with the stated principal value.

Expanding the absolutely convergent Fourier series of β_2 gives

$$\|\mathcal{H}_t C\|_q \leq C_q \sum_{\alpha} |c_2(\alpha)| \|C\|_q, \quad \|\mathcal{D}_t C\|_q \leq \frac{3}{2} \sum_{k \neq 0} k^{-2} \|C\|_q.$$

If $t - t' \in 4^\ell \mathbb{Z}_2$, the frequencies with $q(\alpha) \leq 2^{2\ell}$ cancel. Applying (7.15) to the remaining frequencies proves both operator-continuity estimates. $\square$

Lemma 7.5. *For any finite set $\mathcal{P}$ of odd primes put*

$$B(V) := \prod_{p \text{ odd}, p \notin \mathcal{P}} \beta_p(V_p), \quad \overline{W} := B(V), \quad \overline{D} := \mathcal{D}_t B.$$

For every fixed $q > 0$,

$$\sup_{\mathcal{P}, k, t} \mathbb{E} \left[B(V+k)^q + B(V+k)^{-q} + \overline{D}^q + \overline{D}^{-q} + |\mathcal{H}_t B|^q \right] < \infty.$$

In particular, W, D_t have uniformly bounded positive and negative moments, and h_t has uniformly bounded absolute moments. For $q > 1$ and $t - t' \in 4^\ell \mathbb{Z}_2$,

$$\|h_t - h_{t'}\|_q + \|D_t - D_{t'}\|_q \ll_q 2^{-\ell}. \quad (7.16)$$

Proof. Taylor expansion gives $\mathbb{E}\beta_p^{\pm q} = 1 + O_q(p^{-2})$. Independence and the almost-sure positivity proved above imply

$$\sup_{\mathcal{P}, k} \mathbb{E} B(V+k)^{\pm q} \leq \prod_{p > 2} (1 + C_q p^{-2}) < \infty. \quad (7.17)$$

Higher moments justify passage to the infinite products. Minkowski bounds $\overline{D}$ by the summable weights k^{-2} . For inverse moments choose

$$k_0 := \begin{cases} 1, & t \equiv 0, 1 \pmod{4}, \\ -1, & t \equiv 2, 3 \pmod{4}. \end{cases} \quad (7.18)$$

Then $\beta_2(t + k_0) = 3/2$ and $\overline{D} \geq \frac{3}{2} B(V + k_0)$, so (7.17) also bounds $\mathbb{E}\overline{D}^{-q}$. Lemma 7.4 bounds $\mathcal{H}_t B$. Taking $\mathcal{P} = \emptyset$ proves the assertions for W, D_t, h_t , including (7.16); lower moments follow from Hölder. $\square$

The lemma justifies the principal value in (7.1) and the continuity in t used above. Finally,

$$\mathbb{E} \sum_{k \neq 0} \frac{A_t(k)}{k^2} < \infty,$$

so the $O(k^{-2})$ regularized kernel gives the normal convergence asserted in (7.2).

7.4 Independent prime blocks

We now expose a block of large odd primes. After conditioning on all coordinates outside the block, the contribution of each block prime is independent. Each contribution is small, but the product of their characteristic functions gives strong Fourier decay. The point of this subsection is to show that the interactions involving two or more block primes are small enough to retain rather than discard.

Let Y be large, set

$$\mathcal{P} := \{p : Y < p \leq 2Y\}, \quad N := |\mathcal{P}| \asymp Y/\log Y,$$

and let $\mathcal{F}$ be generated by the odd-prime coordinates outside $\mathcal{P}$. Remove the block from the odd-prime product by writing

$$B(V) := \prod_{\substack{p>2 \\ p \notin \mathcal{P}}} \beta_p(V_p), \quad B_k := B(V+k).$$

Then

$$W(V+k) = B_k \prod_{p \in \mathcal{P}} (1 + \delta_p(V_p + k)).$$

Put

$$b_Y := \sum_{p \in \mathcal{P}} p^{-1} \ll (\log Y)^{-1}, \quad \sigma^2 := \sum_{p \in \mathcal{P}} p^{-2} \asymp (Y \log Y)^{-1}.$$

For $S \subseteq \mathcal{P}$, put

$$W_S := B(V) \prod_{p \in S} \delta_p(V_p), \quad h_S := \mathcal{H}_t W_S, \quad D_S := \mathcal{D}_t W_S.$$

Thus S records exactly which block primes occur in a term. The product expansion is exact and gives

$$h_t = h_\emptyset + \sum_{p \in \mathcal{P}} h_p + \mathcal{R}, \quad \mathcal{R} := \sum_{|S| \geq 2} h_S. \quad (7.19)$$

Conditional on $\mathcal{F}$, the one-prime terms $h_p = h_{\{p\}}$ are independent and centered. The variables appearing in the weight ψ have the simultaneous expansion

$$Z := (D_t, W), \quad \bar{Z} := (\mathcal{D}_t B, B),$$

and

$$\Delta Z := Z - \bar{Z} = \sum_{S \neq \emptyset} (D_S, W_S).$$

Conditional on $\mathcal{F}$, each S -term uses only the coordinates in S .

Lemma 7.6. *For every fixed $q > 1$,*

$$\|\mathcal{R}\|_q \ll_q \sigma^2, \quad \|\Delta Z\|_q \ll_q \sigma. \quad (7.20)$$

Moreover, for every $S \subseteq \mathcal{P}$,

$$\|h_S\|_q + \|D_S\|_q + \|W_S\|_q \ll_q \prod_{p \in S} p^{-1}. \quad (7.21)$$

The coordinates of Z and $\bar{Z}$ are pointwise comparable:

$$e^{-2b_Y} \leq \frac{D_t}{D} \leq e^{b_Y}, \quad e^{-2b_Y} \leq \frac{W}{\bar{W}} \leq e^{b_Y}. \quad (7.22)$$

Proof. Put $P := \prod_{p \in \mathcal{P}} (1 + \delta_p)$ and $S_1 := \sum_{p \in \mathcal{P}} \delta_p$. Since $\sum |\delta_p| \leq b_Y = o(1)$ and $\sum \delta_p^2 \leq \sigma^2$,

$$|P - 1 - S_1| \ll S_1^2 + \sigma^2.$$

In the expansion of $\mathbb{E} S_1^{2r}$, centering removes singleton indices, while for every $j \geq 2$,

$$\sum_{p \in \mathcal{P}} \mathbb{E} |\delta_p|^j \leq Y^{-(j-2)} \sigma^2 \leq \sigma^j.$$

Hence $\|S_1\|_q \ll_q \sigma$ and $\|P - 1 - S_1\|_q \ll_q \sigma^2$. Independence of B and the block, together with Lemmas 7.4 and 7.5, gives

$$\|\mathcal{H}_t(B(P - 1 - S_1))\|_q \ll_q \sigma^2, \quad \|\mathcal{D}_t(B(P - 1))\|_q + \|B(P - 1)\|_q \ll_q \sigma.$$

The same operator bounds applied to $B \prod_{p \in S} \delta_p$ prove (7.21). Finally $e^{-2b_Y} \leq P(V + k) \leq e^{b_Y}$ and positivity of the slope weights imply (7.22). $\square$

To prove contraction, we truncate h_p to nearby poles. The following Fourier estimate controls the omitted poles. For an integer $K \geq 1$, let

$$L_K(\theta) := \sum_{0 < |k| \leq K} \frac{e(k\theta)}{k}, \quad T_K(\theta) := \text{PV} \sum_{|k| > K} \frac{e(k\theta)}{k}.$$

For every $p \geq 2$, $K \geq 1$, and real θ , we have

$$\frac{1}{p} \sum_{a=0}^{p-1} |T_K(\theta + a/p)|^2 \ll K^{-1} + p^{-1}. \quad (7.23)$$

Indeed, the uniformly bounded sine harmonic sums and summation by parts give

$$|T_K(\theta)| \ll \min \left(1, \frac{1}{K \|\theta\|_{\mathbb{R}/\mathbb{Z}}} \right) \quad (\theta \notin \mathbb{Z});$$

at integers the tail is zero. On a shifted p -grid, at most a fixed number of points lie at each successive distance j/p from an integer. Hence the left side of (7.23) is bounded by

$$\frac{C}{p} \left(1 + \sum_{j=1}^p \min \left(1, \frac{p^2}{K^2 j^2} \right) \right) \ll p^{-1} + K^{-1}.$$

Take $K = \lfloor Y/4 \rfloor$, so $2K < p$ throughout the block. Set

$$w_k := \frac{\beta_2(t+k)B_k}{k}, \quad M_K := \sum_{0 < |k| \leq K} |w_k|, \quad X_p := \sum_{0 < |k| \leq K} w_k \delta_p(V_p + k), \quad r_p := h_p - X_p.$$

Lemma 7.7. *Uniformly in t and $p \in \mathcal{P}$,*

$$\mathbb{E}_{\mathcal{F}} \|r_p\|_{L^2(V_p)}^2 \ll p^{-2} Y^{-1}.$$

Proof. Fix first a dyadic frequency α of $\beta_2(t+k)$. Write $\widehat{B}(\theta)$ for the Fourier coefficients of B on the outside coordinates. Since B has no p -coordinate, the pairs $(\theta, a/p^\nu)$ are distinct characters. Parseval gives

$$\begin{aligned} & \left\| \text{PV} \sum_{|k|>K} \frac{e(k\alpha)B(V+k)\delta_p(V_p+k)}{k} \right\|_2^2 \\ &= \sum_{\theta} |\widehat{B}(\theta)|^2 \sum_{\nu \geq 1} \sum_{a \bmod p^\nu}^* p^{-3\nu} |T_K(\alpha + \theta + a/p^\nu)|^2. \end{aligned}$$

The terms with denominator p contribute $O(p^{-2}(K^{-1}+p^{-1}))$ by (7.23); higher p -power denominators contribute $O(p^{-4})$ since T_K is bounded. With $K \asymp Y$ and $p \asymp Y$, their sum is $O(p^{-2}Y^{-1})$. Finally $\|B\|_2 \ll 1$, and Minkowski sums the dyadic frequencies at cost $2 + \sqrt{2}$. $\square$

The truncated term has a variance bounded below on an event determined by the coordinates outside the block. Indeed, $2K < p$, so distinct indices in the truncation are distinct modulo p . The covariance formula (7.14) gives

$$\text{Var}(X_p \mid \mathcal{F}) = (v_p + p^{-3}) \sum w_k^2 - p^{-3} \left(\sum w_k \right)^2. \quad (7.24)$$

Moreover $|X_p| \leq M_K/p$, and at the index in (7.18), $|w_{k_0}| = (3/2)B_{k_0}$. Consequently, if

$$|w_{k_0}| \geq a, \quad M_K \leq M, \quad p \geq CM^2/a^2,$$

then $\text{Var}(X_p \mid \mathcal{F}) \geq c_0 a^2/p^2$. Also, for each fixed q ,

$$\mathbb{E}|w_{k_0}|^{-q} \ll_q 1, \quad \mathbb{E}M_K^q \ll_q (\log(2K))^q. \quad (7.25)$$

The next estimate shows that this variance still gives contraction after the truncation error is restored.

Lemma 7.8. *Suppose centered real X, r satisfy*

$$|X| \leq M/p, \quad \text{Var } X \geq c_0 a^2/p^2, \quad \mathbb{E}r^2 \leq \kappa a^2/p^2.$$

If $TM/p \leq 1/8$ and κ is sufficiently small in terms of c_0 , then

$$|\mathbb{E}e^{iT(X+r)}| \leq \exp(-c_1 T^2 a^2/p^2). \quad (7.26)$$

Independence between X and r is not required.

Proof. Let (X', r') be an independent copy. Since $|T(X - X')| \leq 1/4$, symmetrization gives

$$\|e^{iTX} - e^{iTX'}\|_2^2 = 2\mathbb{E}[1 - \cos(T(X - X'))] \geq cT^2 \text{Var } X.$$

The reverse triangle inequality and $|e^{iu} - e^{iv}| \leq |u - v|$ imply

$$\sqrt{2(1 - |\mathbb{E}e^{iT(X+r)}|^2)} \geq cT\sqrt{\text{Var } X} - 2T\|r\|_2 \geq c'Ta/p$$

for sufficiently small κ . Squaring and using $1 - u \leq e^{-u}$ proves (7.26). $\square$

7.5 Proof of the weighted smoothing estimate

We now combine the preceding estimates. Write $h_t = \ell + \mathcal{R}$, where $\ell := h_\emptyset + \sum_{p \in \mathcal{P}} h_p$ contains the independent one-prime terms and $\mathcal{R}$ contains all interactions. On a suitable outside event, most of the one-prime terms contract the characteristic function. A term supported on $S \subseteq \mathcal{P}$ still leaves all primes outside S available for this contraction. We therefore retain the linear contribution of every interaction term and estimate only the quadratic Taylor remainder in absolute value.

Lemma 7.9. *Let $\mathbf{P}$ be a probability law. Conditionally on a sigma-field $\mathcal{F}$, let $(V_p)_{p \in \mathcal{P}}$ be independent, put $N := |\mathcal{P}|$, and write*

$$\ell = \ell_\emptyset + \sum_{p \in \mathcal{P}} \ell_p,$$

where $\ell_\emptyset$ and the ℓ_p are real-valued, $\ell_\emptyset$ is $\mathcal{F}$ -measurable, and ℓ_p is measurable with respect to $\mathcal{F} \vee \sigma(V_p)$. Suppose that on an event $G \in \mathcal{F}$ there is an $\mathcal{F}$ -measurable set $\mathcal{P}_{\text{good}} \subseteq \mathcal{P}$, of cardinality at least $N/2$, such that

$$\left| \mathbb{E}_{\mathbf{P}}(e^{iT\ell_p} \mid \mathcal{F}) \right| \leq e^{-\gamma} \quad (p \in \mathcal{P}_{\text{good}}).$$

If $Q_S \in L^1(\mathbf{P})$ is measurable with respect to $\mathcal{F} \vee \sigma(V_p : p \in S)$ for every $S \subseteq \mathcal{P}$, then

$$\left| \mathbb{E}_{\mathbf{P}} \left[\mathbf{1}_G e^{iT\ell} \sum_{S \subseteq \mathcal{P}} Q_S \right] \right| \leq e^{-\gamma N/4} \sum_{|S| \leq N/4} \mathbb{E}_{\mathbf{P}} |Q_S| + \sum_{|S| > N/4} \mathbb{E}_{\mathbf{P}} |Q_S|. \quad (7.27)$$

In particular, if $\mathcal{P} \subset (Y, 2Y]$, $N \asymp Y/\log Y$, $\gamma \geq cT^2 a^2/Y^2$, and

$$\mathbb{E}_{\mathbf{P}} |Q_S| \leq C(1 + |T|) \prod_{p \in S} \frac{C_0}{p},$$

then the left side of (7.27) is

$$\ll C(1 + |T|) \left\{ \exp \left(-c' \frac{T^2 a^2}{Y \log Y} \right) + e^{-c'Y} \right\}.$$

Proof. For $|S| \leq N/4$, at least $N/4$ good coordinates lie outside S . Condition first on $\mathcal{F}$ and on the coordinates in S , and use the remaining independent factors. This gives

$$\left| \mathbb{E}_{\mathbf{P}}[\mathbf{1}_G Q_S e^{iT\ell}] \right| \leq e^{-\gamma N/4} \mathbb{E}_{\mathbf{P}} |Q_S|.$$

For larger S , use the trivial bound and sum. Finally, with $b := \sum_{p \in \mathcal{P}} C_0/p \ll 1/\log Y$,

$$\sum_S \prod_{p \in S} \frac{C_0}{p} \leq e^b \ll 1, \quad \sum_{|S| > N/4} \prod_{p \in S} \frac{C_0}{p} \leq \sum_{j > N/4} \frac{b^j}{j!} \leq e^{-c'Y}.$$

This proves both assertions. $\square$

Set $T := |\xi|$ large. We choose

$$Y := T^{7/4}, \quad K := \lfloor Y/4 \rfloor, \quad a := T^{-1/16}, \quad M := T^{1/16} \log(2K). \quad (7.28)$$

The block scale Y makes the interaction remainder small, while the thresholds a and M retain enough variance in each one-prime term. Let G be the $\mathcal{F}$ -measurable event on which $|w_{k_0}| \geq a$, $M_K \leq M$, and

$$\frac{1}{N} \sum_{p \in \mathcal{P}} p^2 \|r_p\|_{L^2(V_p)}^2 \leq \kappa a^2/2.$$

By Lemma 7.7, (7.25), and Markov,

$$\mathbb{P}(G^c) \ll T^{-26/16} + a^{-2}Y^{-1} \ll T^{-13/8}, \quad (7.29)$$

using moments of order 26. On G , at least $N/2$ primes satisfy $\|r_p\|_2^2 \leq \kappa a^2/p^2$. The choice (7.28) ensures $p \geq CM^2/a^2$ and $TM/p \leq 1/8$ for large T , so Lemma 7.8 applies to these primes.

Proof of Proposition 7.2. It is enough to consider $T := |\xi|$ large, since bounded T follows from Lemma 7.5. Use the block and the good event above, and write $h_t = \ell + \mathcal{R}$ as in (7.19).

For admissible ψ , Taylor expansion and (7.22) give

$$\psi(Z)e^{iT\mathcal{R}} = \psi(\bar{Z}) + \nabla\psi(\bar{Z}) \cdot \Delta Z + iT\psi(\bar{Z})\mathcal{R} + \mathcal{E},$$

where, for some fixed m ,

$$|\mathcal{E}| \leq C_\psi \omega(\bar{Z})^m (|\Delta Z|^2 + T|\Delta Z||\mathcal{R}| + T^2|\mathcal{R}|^2).$$

By Hölder, Lemma 7.5, and (7.20),

$$\mathbb{E}|\mathcal{E}| \ll_\psi \sigma^2 + T\sigma^3 + T^2\sigma^4 \ll_\psi \sigma^2 + T^2\sigma^4. \quad (7.30)$$

The retained terms have the supported form required by Lemma 7.9. Namely, put

$$Q_\emptyset := \psi(\bar{Z})$$

and, for $S \neq \emptyset$,

$$Q_S := \nabla\psi(\bar{Z}) \cdot (D_S, W_S) + \mathbf{1}_{\{|S| \geq 2\}} iT\psi(\bar{Z})h_S.$$

By Hölder, Lemmas 7.5 and 7.6, and the polynomial growth of ψ and its derivatives,

$$\mathbb{E}|Q_S| \ll_\psi (1+T) \prod_{p \in S} \frac{C_0}{p}$$

for a fixed absolute C_0 . The contraction estimate above verifies the good-coordinate hypothesis of Lemma 7.9. Since $\|\psi(Z)\|_{13} \ll_\psi 1$, that lemma, (7.30), and Hölder on G^c give

$$|\mathbb{E}[\psi(Z)e^{iT h_t}]| \ll_\psi (1+T) \left(e^{-cT^2 a^2/(Y \log Y)} + e^{-cY} \right) + \sigma^2 + T^2\sigma^4 + \mathbb{P}(G^c)^{12/13} \ll_\psi T^{-3/2}.$$

Indeed, the exponential terms are negligible, $\sigma^2 \ll T^{-7/4}$, $T^2\sigma^4 \ll T^{-3/2}$, and (7.29) gives $\mathbb{P}(G^c)^{12/13} \ll T^{-3/2}$. This proves (7.6); the argument is identical for either sign of ξ . $\square$

8 Positivity of the density

Proposition 8.1. *The continuous density of h_0 is strictly positive at every real point:*

$$f_1(0, y) > 0 \quad (y \in \mathbb{R}).$$

In particular $\nu_{0,0} \neq 0$.

Write $h := h_0$ and $f(y) := f_1(0, y)$. We combine uniform smoothing under finite changes of measure with a limit that moves one pole contribution, then average the displacement. Uniform smoothing allows positivity to pass through the limit.

8.1 A class of tilted product measures

All prime products in this section are over odd primes. Fix a nonzero integer k_* , a number $0 < \vartheta_0 < 1$, and a finite set I of odd primes. For $0 \leq \vartheta_p \leq \vartheta_0$ define

$$\frac{dQ}{d\mathbb{P}}(V) := \prod_{p \in I} (1 + \vartheta_p p \delta_p(V_p + k_*)), \quad \sum_{p \in I} \frac{\vartheta_p}{p} \leq L. \quad (8.1)$$

These are probability measures with independent coordinates. Each local density lies between $1 - \vartheta_0$ and $1 + \vartheta_0$. For a fixed finite I , the full density is bounded above and below by positive constants; these bounds need not be uniform in I . Use $\vartheta_p = 0$ off I and write Q_p for the local law. We write $\|X\|_{q,Q} := (\mathbb{E}_Q |X|^q)^{1/q}$.

Let $\mathcal{A}$ be the algebra of integer sequences with absolutely summable rational Fourier coefficients, with norm the sum of their absolute values. The covariance formula gives

$$\begin{aligned} m_p(k) &:= \mathbb{E}_{Q_p} \delta_p(V_p + k) = \vartheta_p p C_p(k - k_*), \\ C_p(a) &:= \sum_{\nu \geq 1} p^{-3\nu} c_{p^\nu}(a), \end{aligned} \quad \|m_p\|_{\mathcal{A}} = \vartheta_p p v_p = \frac{\vartheta_p}{p+1}.$$

Set $\mu_p(k) := 1 + m_p(k)$ and $\xi_p(V_p, k) := \delta_p(V_p + k) - m_p(k)$. For any set of indices E ,

$$\left\| \prod_{p \in E} \mu_p \right\|_{\mathcal{A}} \leq e^L.$$

Thus the deterministic mean factors have a uniform Fourier norm even when the full Radon–Nikodym derivative is very large.

Lemma 8.2. *For $F \in \mathcal{A}$, define*

$$\mathcal{T}_F C := \text{PV} \sum_{k \neq 0} \frac{\beta_2(k) F(k) C(V + k)}{k}, \quad \mathcal{T}_{F, > K} C := \text{PV} \sum_{|k| > K} \frac{\beta_2(k) F(k) C(V + k)}{k}.$$

Under Haar measure,

$$\begin{aligned} \|\mathcal{T}_F C\|_2 &\leq C \|F\|_{\mathcal{A}} \|C\|_2, \\ \|\mathcal{T}_{F, > K} (C \delta_p)\|_2^2 &\leq C \|F\|_{\mathcal{A}}^2 \|C\|_2^2 p^{-2} (K^{-1} + p^{-1}), \end{aligned} \quad (8.2)$$

where the second assertion assumes that C is independent of V_p .

Proof. Expand F in its absolutely summable Fourier series. Each frequency modulates the operator in Lemma 7.4 and shifts the argument of T_K in the Parseval calculation of Lemma 7.7. The grid bound (7.23) is uniform in this shift; the higher conductors cost $O(p^{-4})$. Minkowski sums the frequencies at cost $\|F\|_{\mathcal{A}}$ in the norm. $\square$

Lemma 8.3. *Let $J := \{p : Y < p \leq 2Y\}$, and let $\mathcal{F}$ be generated by the coordinates outside J . Put $B(V + k) := \prod_{p \notin J} \beta_p(V_p + k)$. The exact conditional centered expansion is*

$$h = \sum_{S \subseteq J} h_S^Q, \quad h_S^Q := \text{PV} \sum_{k \neq 0} \frac{\beta_2(k) B(V + k)}{k} \prod_{p \in J \setminus S} \mu_p(k) \prod_{p \in S} \xi_p(V_p, k).$$

There is an absolute constant $C_0 \geq 1$ such that, with constants depending only on L, ϑ_0 ,

$$\|h_S^Q\|_{2,Q} \leq C_L \prod_{p \in S} \frac{C_0}{p}, \quad \left\| \sum_{|S| \geq 2} h_S^Q \right\|_{2,Q} \ll_L \sum_{p \in J} p^{-2}. \quad (8.3)$$

For every fixed $q > 0$, $\mathbb{E}_Q B(V+k)^{\pm q} \ll_{q,L} 1$ uniformly in k, J, I . If X_p truncates $h_{\{p\}}^Q$ to $0 < |k| \leq K$, then

$$\mathbb{E}_Q |h_{\{p\}}^Q - X_p|^2 \ll_L p^{-2} (K^{-1} + p^{-1}). \quad (8.4)$$

The first estimate in (8.3) also holds for any finite J , with the same constants.

Proof. Expand the outside tilted primes as well. For $S \subseteq J$ and $T \subseteq I \setminus J$, put

$$E := S \cup T, \quad B_0 := \prod_{p \notin I \cup J} \beta_p, \quad F_{S,T} := \prod_{p \in (I \cup J) \setminus E} \mu_p.$$

With the commuting orthogonal projections $\Pi_p := 1 - \mathbb{E}_{Q_p}$, the term having centered support E is

$$h_{S,T}^Q := \left(\prod_{p \in E} \Pi_p \right) \mathcal{T}_{F_{S,T}} \left(B_0 \prod_{p \in E} \delta_p \right). \quad (8.5)$$

This follows first for finite sums and then in L^2 . The input depends only on E and Haar coordinates, so comparison with Haar costs only $(1 + \vartheta_0)^{|E|}$. Since $\|F_{S,T}\|_{\mathcal{A}} \leq e^L$, projection and Lemma 8.2 give

$$\|h_{S,T}^Q\|_{2,Q}^2 \leq C e^{2L} \|B_0\|_2^2 \prod_{p \in E} (1 + \vartheta_0) v_p.$$

Different centered supports are orthogonal. Summing over T costs the bounded product $\prod_{p \in I \setminus J} (1 + \vartheta_0) v_p$, and hence

$$\|h_S^Q\|_{2,Q}^2 \leq C_L \prod_{p \in S} C_0^2 p^{-2}, \quad \left\| \sum_{|S| \geq 2} h_S^Q \right\|_{2,Q}^2 \leq C_L \sum_{|S| \geq 2} \prod_{p \in S} C_0^2 p^{-2} \ll_L \left(\sum_{p \in J} p^{-2} \right)^2.$$

This also justifies the expansions and applies to any finite J .

For the moments, Taylor expansion gives

$$\mathbb{E}_{Q_p} \beta_p (V_p + k)^{\pm q} = 1 \pm q m_p(k) + O_q(p^{-2}), \quad \sum_p |m_p(k)| \leq L.$$

Multiplication over independent coordinates proves the claimed bounds. For the tail use (8.5) with $S = \{p\}$ and replace $\mathcal{T}_F$ by $\mathcal{T}_{F, > K}$. Before projection the input $C = B_0 \prod_{q \in T} \delta_q$ is independent of V_p , so (8.2) gives

$$\|\mathcal{T}_{F_{\{p\}, T, > K}}(C \delta_p)\|_{2, \mathbb{P}}^2 \ll e^{2L} \|B_0\|_2^2 \prod_{q \in T} v_q p^{-2} (K^{-1} + p^{-1}).$$

Comparison on $T \cup \{p\}$, contraction of the projections, and orthogonal summation over T prove (8.4). $\square$

Proposition 8.4. *All the measures (8.1) satisfy*

$$|\mathbb{E}_Q e^{i\xi h}| \ll_{L, \vartheta_0} (1 + |\xi|)^{-3/2}. \quad (8.6)$$

In particular their h -densities are continuous and have a common integrable Fourier majorant. The bound does not involve $|I|$ or the supremum of $dQ/d\mathbb{P}$.

Proof. Put $T := |\xi|$ and assume that T is large; bounded T is trivial. Use the block $J = (Y, 2Y] \cap \{\text{odd primes}\}$ and parameters (7.28). Write $h = \ell_Q + \mathcal{R}_Q$, with $\ell_Q := h_\emptyset^Q + \sum_{p \in J} h_{\{p\}}^Q$. The one-prime truncations are

$$X_p := \sum_{0 < |k| \leq K} w_{p,k} \xi_p(V_p, k), \quad w_{p,k} := \frac{\beta_2(k) B(V+k)}{k} \prod_{q \in J \setminus \{p\}} \mu_q(k).$$

The mean products lie between e^{-2b_Y} and e^{b_Y} . On the outside event

$$\frac{3}{2} B(V+1) \geq 2a, \quad \sum_{0 < |k| \leq K} \frac{\beta_2(k) B(V+k)}{|k|} \leq M/4,$$

we therefore have $|w_{p,1}| \geq a$ and $|X_p| \leq M/p$. Since $dQ_p/d\mathbb{P}_p \geq 1 - \vartheta_0$,

$$\text{Var}_{Q_p} X \geq (1 - \vartheta_0) \text{Var}_{\mathbb{P}_p} X.$$

Apply this to $\sum w_{p,k} \delta_p(V_p + k)$ and use (7.24): $\text{Var}_{Q_p} X_p \geq c_{\vartheta_0} a^2/p^2$. Lemma 8.3 supplies the uniform moment and tail bounds. The argument of (7.29) gives an outside event G with

$$Q(G^c) \ll_{L, \vartheta_0} T^{-13/8}, \quad (8.7)$$

on which at least half the block primes satisfy Lemma 7.8.

Apply Lemma 7.9 under the law $\mathbb{P} = Q$, with $\ell = \ell_Q$. The expansion

$$e^{iT\mathcal{R}_Q} = 1 + iT\mathcal{R}_Q + \mathcal{E}_Q, \quad |\mathcal{E}_Q| \leq \frac{1}{2} T^2 |\mathcal{R}_Q|^2,$$

has the required supported form: take $Q_\emptyset = 1$, $Q_S = 0$ for $|S| = 1$, and $Q_S = iT h_S^Q$ for $|S| \geq 2$. By (8.3), uniformly in I ,

$$\mathbb{E}_Q |Q_S| \ll_{L, \vartheta_0} (1 + T) \prod_{p \in S} \frac{C_0}{p}, \quad \mathbb{E}_Q |\mathcal{E}_Q| \ll_{L, \vartheta_0} T^2 \sigma^4.$$

The contraction above verifies the good-coordinate hypothesis. Lemma 7.9 and (8.7) therefore give

$$|\mathbb{E}_Q e^{iT h}| \ll_{L, \vartheta_0} (1 + T) \left(e^{-cT^2 a^2 / (Y \log Y)} + e^{-cY} \right) + T^2 \sigma^4 + Q(G^c) \ll_{L, \vartheta_0} T^{-3/2}.$$

This proves (8.6); the estimate is uniform in I . $\square$

8.2 Changing one pole contribution in the limit

Fix $A > 1$ and $0 \leq \vartheta \leq \vartheta_0$. Let $Q_{P, \vartheta}$ use $I_P := \{p : P < p \leq P^A\}$ and $\vartheta_p := \vartheta$ in (8.1). Mertens' prime sum formula gives $\sum_{p \in I_P} p^{-1} \rightarrow \log A$. Define the signed pole contribution

$$a_*(V) := \frac{\beta_2(k_*)}{k_*} W(V + k_*).$$

Proposition 8.5. *As $P \rightarrow \infty$,*

$$\text{Law}_{Q_{P,\vartheta}}(h) \implies \text{Law}_{\mathbb{P}}(h + (A^\vartheta - 1)a_*).$$

Convergence of characteristic functions is uniform for $\vartheta \in [0, \vartheta_0]$ at each fixed frequency.

Proof. Put $B_P(V) := \prod_{p \notin I_P} \beta_p(V_p)$ and $\sigma_P^2 := \sum_{p \in I_P} p^{-2} = o(1)$. The outside coordinates have Haar law. The conditional mean is

$$\bar{h}_{P,\vartheta} := \text{PV} \sum_{k \neq 0} \frac{\beta_2(k) B_P(V + k)}{k} M_{P,\vartheta}(k), \quad M_{P,\vartheta}(k) := \prod_{p \in I_P} (1 + \vartheta p C_p(k - k_*)).$$

The centered-support estimate and orthogonality in Lemma 8.3, with $J = I_P$, give

$$\|h - \bar{h}_{P,\vartheta}\|_{2,Q_{P,\vartheta}} \ll_{A,\vartheta_0} \sigma_P. \quad (8.8)$$

The spatial concentration of the mean follows from the Ramanujan-sum identity

$$p C_p(a) = -p^{-2} + (1 - p^{-3}) \sum_{\nu \geq 1} p^{1-2\nu} \mathbf{1}_{\{p^\nu | a\}}.$$

Consequently

$$M_{P,\vartheta}(k) = \kappa_{P,\vartheta} \sum_d a_\vartheta(d) \mathbf{1}_{\{d | k - k_*\}}, \quad \kappa_{P,\vartheta} := \prod_{p \in I_P} (1 - \vartheta/p^2),$$

where a_ϑ is multiplicative on I_P -smooth integers and

$$a_\vartheta(1) := 1, \quad a_\vartheta(p^\nu) := \frac{\vartheta(1 - p^{-3})p^{1-2\nu}}{1 - \vartheta/p^2} \geq 0.$$

Its total mass is uniformly bounded, because

$$\kappa_{P,\vartheta} \sum_d a_\vartheta(d) = M_{P,\vartheta}(k_*) = \prod_{p \in I_P} (1 + \vartheta/(p+1)) \longrightarrow A^\vartheta \quad (8.9)$$

uniformly in ϑ . Also $\kappa_{P,\vartheta} \rightarrow 1$ uniformly.

If $d > 2|k_*|$, the restricted Hilbert sum satisfies

$$\left\| \text{PV} \sum_{\substack{k \neq 0 \\ k \equiv k_* \pmod{d}}} \frac{\beta_2(k) B_P(V + k)}{k} - \frac{\beta_2(k_*) B_P(V + k_*)}{k_*} \right\|_2 \ll d^{-1}.$$

To see this, remove $k = k_*$ and write $k = k_* + dj$. The resulting operator is d^{-1} times a modulated Hilbert transform along $V \mapsto V + d$, with kernel $(j + k_*/d)^{-1}$. Its difference from $1/j$ has ℓ^1 norm $O(|k_*|/d)$. The Fourier norm of $\beta_2(k_* + dj)$ is at most $2 + \sqrt{2}$, proving the bound. The same argument is uniform in symmetric cutoffs $|k| \leq L$ with $L \geq |k_*|$, so $\sum_d a_\vartheta(d) \ll_A 1$ permits interchange of the divisor series and principal value in L^2 . Since every nontrivial divisor exceeds P , the total error is $O(\sum_{d>1} a_\vartheta(d)/d) = O_A(P^{-1})$. Hence

$$\bar{h}_{P,\vartheta} = \kappa_{P,\vartheta} \mathcal{H}_0 B_P + (M_{P,\vartheta}(k_*) - \kappa_{P,\vartheta}) \frac{\beta_2(k_*) B_P(V + k_*)}{k_*} + O_{L^2}(P^{-1}).$$

All constants may depend on fixed A, k_*, ϑ_0 and are uniform in ϑ . Under Haar, $W = B_P P_{I_P}$, where $P_{I_P} := \prod_{p \in I_P} \beta_p(V_p)$, and the two factors are independent. Hence

$$\|B_P - W\|_2^2 = \mathbb{E} B_P^2 \mathbb{E} (P_{I_P} - 1)^2 = \mathbb{E} B_P^2 \left(\prod_{p \in I_P} (1 + v_p) - 1 \right) \ll \sigma_P^2.$$

The Hilbert bound and (8.9) now give

$$\sup_{\vartheta \in [0, \vartheta_0]} \|\bar{h}_{P, \vartheta} - (h + (A^\vartheta - 1)a_*)\|_{2, \mathbb{P}} \longrightarrow 0.$$

The conditional mean depends only on outside coordinates, which have Haar law under both measures. Combining this limit with (8.8) proves convergence of characteristic functions, uniformly in ϑ at each fixed frequency. $\square$

8.3 Averaging displacements forces positivity

Proof of Proposition 8.1. Fix $y \in \mathbb{R}$ and choose $\varepsilon \in \{1, -1\}$ with $\mathbb{P}(\varepsilon(y - h) > 0) > 0$, possible since h has a density. Use $k_* := 1$ for $\varepsilon = 1$ and $k_* := -2$ otherwise. Then $a_* = \varepsilon a$, where $a := |a_*| > 0$ almost surely and $\mathbb{E} a^{-1} < \infty$ by Lemma 7.5. Choose $M < \infty$ with $\mathbb{P}(0 < \varepsilon(y - h)/a < M) > 0$ and a smooth probability density r , supported on $[0, M]$ and positive on $(0, M)$. For independent U with density r , $h + \varepsilon U a$ has continuous density

$$g(x) := \mathbb{E} \left[a^{-1} r(\varepsilon(x - h)/a) \right], \quad g(y) > 0.$$

Continuity follows from $\mathbb{E} a^{-1} < \infty$.

Take $\vartheta_0 := 1/2$, $A := (1 + M)^2$, and $\vartheta(u) := \log(1 + u)/\log A$. Let $\bar{Q}_P := \int_0^M r(u) Q_{P, \vartheta(u)} du$. Its h -law converges weakly to $g(x) dx$ by Proposition 8.5. Their budgets are uniformly bounded, so Proposition 8.4 supplies a common integrable Fourier majorant. Fourier inversion gives

$$\|g_P - g\|_\infty \leq (2\pi)^{-1} \int_{\mathbb{R}} |\hat{g}_P - \hat{g}| \longrightarrow 0.$$

Fix a finite P with $g_P(y) > 0$. Set $C_P := (3/2)^{|I_P|}$. Since $d\bar{Q}_P/d\mathbb{P} \leq C_P < \infty$, we have $g_P \leq C_P f$ almost everywhere, hence everywhere by continuity. Therefore

$$f(y) \geq C_P^{-1} g_P(y) > 0.$$

No uniform bound on C_P is needed. $\square$

Declaration on the use of artificial intelligence

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